Schedule Document

General

This package contains the contextual and assembly model for the function defined in the root class.

Schedule contextual model

General

This package contains all the artefacts for the Schedule Contextual Document.

!# shows the diagram.

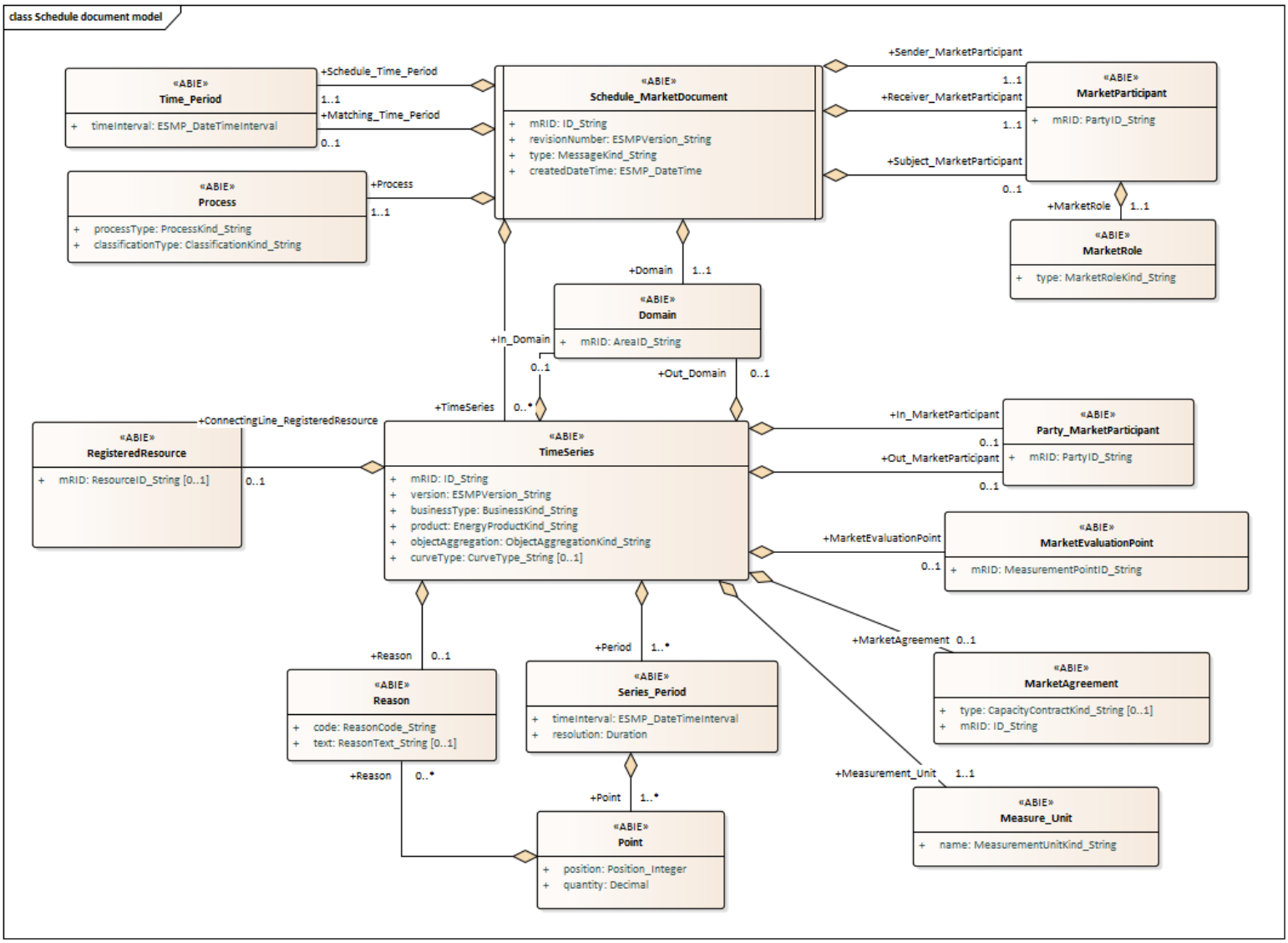


Figure !# - diagram Schedule document model

(ABIE) Domain

A domain covering a number of related objects, such as market balance area, grid area, borders etc.

!# shows all attributes of Domain.

Table !# - Attributes of Schedule contextual model::Domain

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	AreaID_String	The unique identification of the domain.

(ABIE) MarketAgreement

A formal agreement between two parties defining the terms and conditions for a set of services. The specifics of the services are, in turn, defined via one or more service agreements.

!# shows all attributes of MarketAgreement.

Table !# - Attributes of Schedule contextual model::MarketAgreement

Attribute name	mult.	Attribute type	Description
type (ABIE)	0..1	CapacityContractKind_String	The specification of the kind of the agreement, e.g. long term, daily contract.
mRID (ABIE)	1..1	ID_String	The unique identification of the agreement.

(ABIE) MarketEvaluationPoint

The location where one or more products are measured. This may be a physical or virtual location.

!# shows all attributes of MarketEvaluationPoint.

Table !# - Attributes of Schedule contextual model::MarketEvaluationPoint

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	MeasurementPointID_String	A unique identification of the measurement point.

(ABIE) MarketParticipant

The identification of the party participating in energy market business processes.

!# shows all attributes of MarketParticipant.

Table !# - Attributes of Schedule contextual model::MarketParticipant

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	PartyID_String	The identification of a party in the energy market.

!# shows all association ends of MarketParticipant with other classes.

Table !# - Association ends of Schedule contextual model::MarketParticipant with other classes

mult.	mult.	Class type name	Description
MarketRole (ABIE)	[1..1]	MarketRole	The role associated with a MarketParticipant. Association Based On : ESMPClasses:: MarketParticipant.[] ----- ESMPClasses:: MarketRole.MarketRole[0..1]

(ABIE) MarketRole

The identification of the intended behaviour of a market participant played within a given business process.

!# shows all attributes of MarketRole.

Table !# - Attributes of Schedule contextual model::MarketRole

Attribute name	mult.	Attribute type	Description
type (ABIE)	1..1	MarketRoleKind_String	The identification of the role played by a market player.

(ABIE) Measure_Unit

A particular quantity, defined and adopted by convention, with which other quantities of the same kind are compared in order to express their magnitudes relative to that quantity.

!# shows all attributes of Measure_Unit.

Table !# - Attributes of Schedule contextual model::Measure_Unit

Attribute name	mult.	Attribute type	Description
name (ABIE)	1..1	MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20).

(ABIE) Party_MarketParticipant

The identification of the party participating in energy market business processes.

!# shows all attributes of Party_MarketParticipant.

Table !# - Attributes of Schedule contextual model::Party_MarketParticipant

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	PartyID_String	The identification of a party in the energy market.

(ABIE) Point

The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of Schedule contextual model::Point

Attribute name	mult.	Attribute type	Description
position (ABIE)	1..1	Position_Integer	A sequential value representing the relative position within a given time interval.
quantity (ABIE)	1..1	Decimal	The principal quantity identified for a point.

(ABIE) Process

A formal identification of the business process in which a flow of information is exchanged.

!# shows all attributes of Process.

Table !# - Attributes of Schedule contextual model::Process

Attribute name	mult.	Attribute type	Description
processType (ABIE)	1..1	ProcessKind_String	The identification of the nature of process that the document addresses.
classificationType (ABIE)	1..1	ClassificationKind_String	The classification mechanism used to group a set of objects together within a business process. The grouping may be of a detailed or a summary nature.

(ABIE) Reason

The motivation of an act.

!# shows all attributes of Reason.

Table !# - Attributes of Schedule contextual model::Reason

Attribute name	mult.	Attribute type	Description
code (ABIE)	1..1	ReasonCode_String	The motivation of an act in coded form.
text (ABIE)	0..1	ReasonText_String	The textual explanation corresponding to the reason code.

(ABIE) RegisteredResource

A resource that is registered through the market participant registration system. Examples include generating unit, load, and non-physical generator or load.

!# shows all attributes of RegisteredResource.

Table !# - Attributes of Schedule contextual model::RegisteredResource

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	0..1	ResourceID_String	The unique identification of a resource. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

(ABIE) Schedule_MarketDocument

A schedule document provides the position of a party or a domain related to some market information; it includes a set of time series.
An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of Schedule_MarketDocument.

Table !# - Attributes of Schedule contextual model::Schedule_MarketDocument

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ID_String	The unique identification of the document being exchanged within a business process flow.
revisionNumber (ABIE)	1..1	ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
type (ABIE)	1..1	MessageKind_String	The coded type of a document. The document type describes the principal characteristic of the document.
createdDateTime (ABIE)	1..1	ESMP_DateTime	The date and time of the creation of the document.

!# shows all association ends of Schedule_MarketDocument with other classes.

Table !# - Association ends of Schedule contextual model::Schedule_MarketDocument with other classes

mult.	mult.	Class type name	Description
Schedule_Time_Period (ABIE)	[1..1]	Time_Period	This information provides the start and end date and time of the schedule time interval. All time intervals for the time series in the document shall be within the total time interval for the schedule. The receiver will discard any time intervals outside the schedule period. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Time_Period.Period[0..*] This information provides the start and end date and time of the period to be matched. The matching period start date and time shall begin at the start of the schedule time interval or be within the bounds of the schedule time interval. The matching period end date and time

Matching_Time_Period (ABIE)	[0..1] Time_Period	shall be the same as that of the schedule time interval. It is this period that is being presented for matching. The period prior to the matching period is generally considered to be historical data and should correspond to the information received in previous transmissions. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Time_Period.Period[0..*]
Sender_MarketParticipant (ABIE)	[1..1] MarketParticipant	Document owner. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]
Receiver_MarketParticipant (ABIE)	[1..1] MarketParticipant	Document recipient. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]
Subject_MarketParticipant (ABIE)	[0..1] MarketParticipant	The party that is the subject of the documents time series. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]
Process (ABIE)	[1..1] Process	The process dealt with in the document. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Process.Process[0..*]
Domain (ABIE)	[1..1] Domain	The identification of the domain that is covered in the schedule document. It is in general the market balance area that is the subject of the schedule plan. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Domain.Domain[0..1]

(ABIE) Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of Schedule contextual model::Series_Period

Attribute name	mult.	Attribute type	Description
timeInterval (ABIE)	1..1	ESMP_DateTimeInterval	The start and end time of the period.
resolution (ABIE)	1..1	Duration	The definition of the number of units of time that compose an individual step within a period.

(ABIE) Time_Period

The identification of a time interval.

!# shows all attributes of Time_Period.

Table !# - Attributes of Schedule contextual model::Time_Period

Attribute name	mult.	Attribute type	Description
timeInterval (ABIE)	1..1	ESMP_DateTimeInterval	The start and end date and time for a given interval.

(ABIE) TimeSeries

A set of time-ordered quantities being exchanged in relation to a product.

!# shows all attributes of TimeSeries.

Table !# - Attributes of Schedule contextual model::TimeSeries

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ID_String	A unique identification of the time series.
version (ABIE)	1..1	ESMPVersion_String	The identification of the version of the time series.
businessType (ABIE)	1..1	BusinessKind_String	The identification of the nature of the time series.
product (ABIE)	1..1	EnergyProductKind_String	The identification of the nature of an energy product such as power, energy, reactive power, etc.
objectAggregation (ABIE)	1..1	ObjectAggregationKind_String	The identification of the object (party, domain, etc.) that is the common denominator used to aggregate a time series.
curveType (ABIE)	0..1	CurveType_String	The identification of the coded representation of the type of curve being described.

!# shows all association ends of TimeSeries with other classes.

Table !# - Association ends of Schedule contextual model::TimeSeries with other classes

mult.	mult.	Class type name	Description
ConnectingLine_RegisteredResource (ABIE)	[0..1]	RegisteredResource	The identification of a resource associated with a TimeSeries. Association Based On : ESMPClasses:: RegisteredResource.RegisteredResource[0..*] ---- ESMPClasses:: TimeSeries.[]
Measurement_Unit (ABIE)	[1..1]	Measure_Unit	The unit of measurement used for the quantities expressed within the time series. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Measure_Unit.Measurement_Unit[0..*]
MarketEvaluationPoint (ABIE)	[0..1]	MarketEvaluationPoint	The identification of the location where one or more products are metered. This may be one physical location or the combination of several points together. The identification of a measurement point associated with a TimeSeries. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: MarketEvaluationPoint.MarketEvaluationPoint[0..*]
Out_MarketParticipant (ABIE)	[0..1]	Party_MarketParticipant	The identification of the party taking the product out of the out area. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]

In_MarketParticipant (ABIE)	[0..1] Party_MarketParticipant	The identification of the party putting the product into the in area. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]
In_Domain (ABIE)	[0..1] Domain	The area where the product is being delivered. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Domain.Domain[0..*]
Out_Domain (ABIE)	[0..1] Domain	The area where the product is being extracted. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Domain.Domain[0..*]
Reason (ABIE)	[0..1] Reason	At the TimeSeries level the reason code is used to enable processing of the reason text which, depending on market conditions, should be provided in intra day trading. In this context only one reason code has been defined (A48, modification reason). No other codes are permitted. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Reason.Reason[0..*]
MarketAgreement (ABIE)	[0..1] MarketAgreement	The identification of an agreement associated with a time series. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: MarketAgreement.MarketAgreement[0..*]

Schedule assembly model

General

This package contains the model that has been assembled from the contextual model. All classes with relations [1..1] or [0..1] have had their attributes integrated into their parent class.

!# shows the diagram.

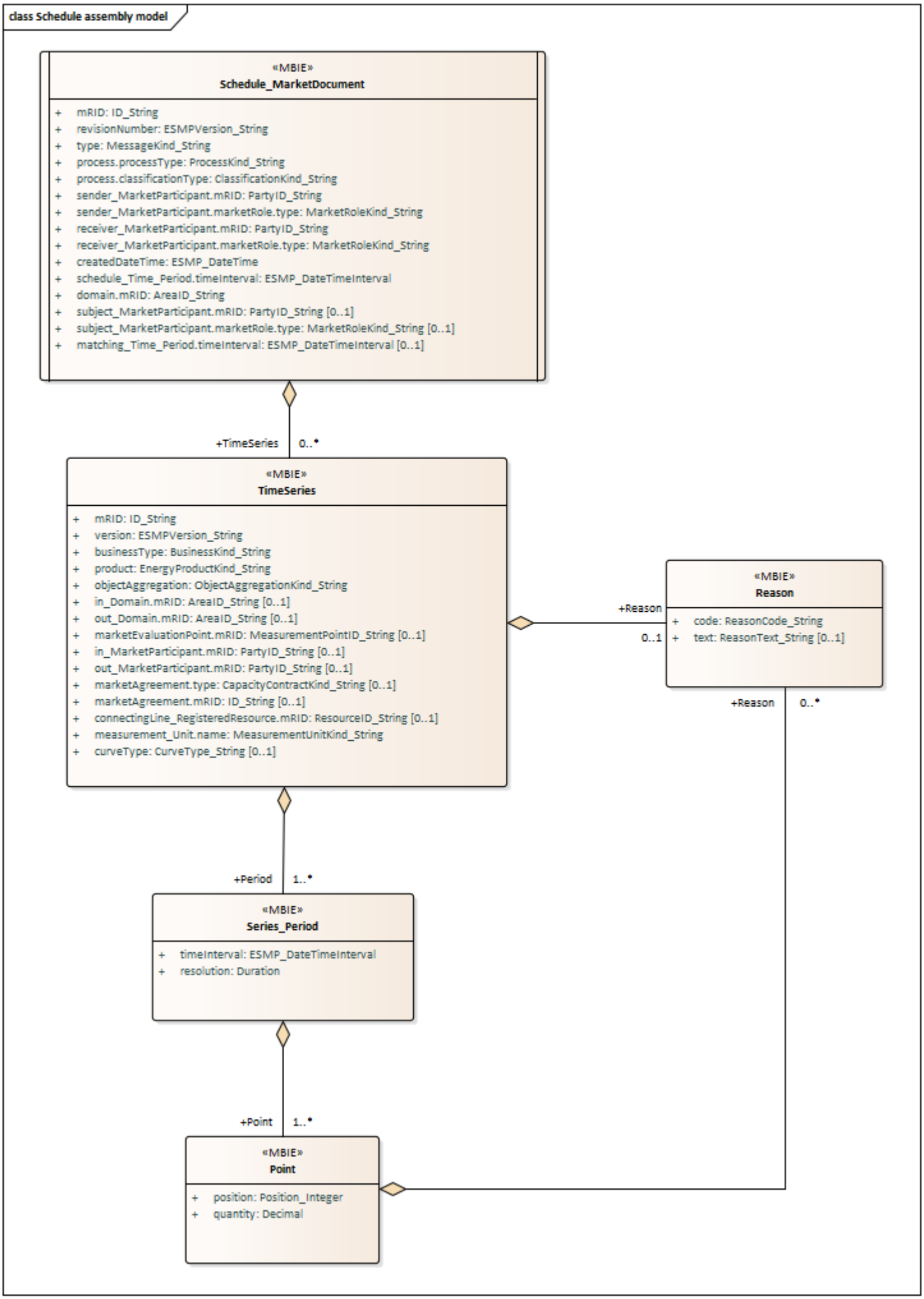


Figure !# - diagram Schedule assembly model

(MBIE) Point

The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of Schedule assembly model::Point

Attribute name	mult.	Attribute type	Description
position (MBIE)	1..1	Position_Integer	A sequential value representing the relative position within a given time interval.
quantity (MBIE)	1..1	Decimal	The principal quantity identified for a point.

(MBIE) Reason

The motivation of an act.

!# shows all attributes of Reason.

Table !# - Attributes of Schedule assembly model::Reason

Attribute name	mult.	Attribute type	Description
code (MBIE)	1..1	ReasonCode_String	The motivation of an act in coded form.

text (MBIE) 0..1 ReasonText_String The textual explanation corresponding to the reason code.

(MBIE) Schedule_MarketDocument

A schedule document provides the position of a party or a domain related to some market information; it includes a set of time series.
An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of Schedule_MarketDocument.

Table !# - Attributes of Schedule assembly model::Schedule_MarketDocument

Attribute name	mult.	Attribute type	Description
mRID (MBIE)	1..1	ID_String	The unique identification of the document being exchanged within a business process flow.
revisionNumber (MBIE)	1..1	ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
type (MBIE)	1..1	MessageKind_String	The coded type of a document. The document type describes the principal characteristic of the document.
process.processType (MBIE)	1..1	ProcessKind_String	The identification of the nature of process that the document addresses. --- The process dealt with in the document.
process.classificationType (MBIE)	1..1	ClassificationKind_String	The classification mechanism used to group a set of objects together within a business process. The grouping may be of a detailed or a summary nature. --- The process dealt with in the document.
sender_MarketParticipant.mRID (MBIE)	1..1	PartyID_String	The identification of a party in the energy market. --- Document owner.
sender_MarketParticipant.marketRole.type (MBIE)	1..1	MarketRoleKind_String	The identification of the role played by a market player. --- Document owner.
receiver_MarketParticipant.mRID (MBIE)	1..1	PartyID_String	--- The role associated with a MarketParticipant. The identification of a party in the energy market. --- Document recipient.
receiver_MarketParticipant.marketRole.type			The identification of the role played by a market player.

(MBIE)	1..1	MarketRoleKind_String	--- Document recipient. --- The role associated with a MarketParticipant.
createdDateTime (MBIE)	1..1	ESMP_DateTime	The date and time of the creation of the document. The start and end date and time for a given interval.
schedule_Time_Period.timeInterval (MBIE)	1..1	ESMP_DateTimeInterval	--- This information provides the start and end date and time of the schedule time interval. All time intervals for the time series in the document shall be within the total time interval for the schedule. The receiver will discard any time intervals outside the schedule period.
domain.mRID (MBIE)	1..1	AreaID_String	The unique identification of the domain. --- The identification of the domain that is covered in the schedule document. It is in general the market balance area that is the subject of the schedule plan.
subject_MarketParticipant.mRID (MBIE)	0..1	PartyID_String	The identification of a party in the energy market. --- The party that is the subject of the documents time series.
subject_MarketParticipant.marketRole.type (MBIE)	0..1	MarketRoleKind_String	The identification of the role played by a market player. --- The party that is the subject of the documents time series. --- The role associated with a MarketParticipant.
matching_Time_Period.timeInterval (MBIE)	0..1	ESMP_DateTimeInterval	The start and end date and time for a given interval. --- This information provides the start and end date and time of the period to be matched. The matching period start date and time shall begin at the start of the schedule time interval or be within the bounds of the schedule time interval. The matching period end date and time shall be the same as that of the schedule time interval. It is this period that is being presented for matching.

			The period prior to the matching period is generally considered to be historical data and should correspond to the information received in previous transmissions.

(MBIE) Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of Schedule assembly model::Series_Period

Attribute name	mult.	Attribute type	Description
timeInterval (MBIE)	1..1	ESMP_DateTimeInterval	The start and end time of the period.
resolution (MBIE)	1..1	Duration	The definition of the number of units of time that compose an individual step within a period.

(MBIE) TimeSeries

A set of time-ordered quantities being exchanged in relation to a product.

!# shows all attributes of TimeSeries.

Table !# - Attributes of Schedule assembly model::TimeSeries

Attribute name	mult.	Attribute type	Description
mRID (MBIE)	1..1	ID_String	A unique identification of the time series.
version (MBIE)	1..1	ESMPVersion_String	The identification of the version of the time series.
businessType (MBIE)	1..1	BusinessKind_String	The identification of the nature of the time series.
product (MBIE)	1..1	EnergyProductKind_String	The identification of the nature of an energy product such as power, energy, reactive power, etc.
objectAggregation (MBIE)	1..1	ObjectAggregationKind_String	The identification of the object (party, domain, etc.) that is the common denominator used to aggregate a time series.
in_Domain.mRID (MBIE)	0..1	AreaID_String	The unique identification of the domain. --- The area where the product is being delivered.
out_Domain.mRID (MBIE)	0..1	AreaID_String	The unique identification of the domain. --- The area where the product is being extracted.
marketEvaluationPoint.mRID (MBIE)	0..1	MeasurementPointID_String	A unique identification of the measurement point. --- The identification of the location where one or more products are metered. This may be one physical location

		or the combination of several points together. The identification of a measurement point associated with a TimeSeries.
in_MarketParticipant.mRID (MBIE)	0..1 PartyID_String	The identification of a party in the energy market. --- The identification of the party putting the product into the in area.
out_MarketParticipant.mRID (MBIE)	0..1 PartyID_String	The identification of a party in the energy market. --- The identification of the party taking the product out of the out area.
marketAgreement.type (MBIE)	0..1 CapacityContractKind_String	The specification of the kind of the agreement, e.g. long term, daily contract. --- The identification of an agreement associated with a time series.
marketAgreement.mRID (MBIE)	0..1 ID_String	The unique identification of the agreement. --- The identification of an agreement associated with a time series.
connectingLine_RegisteredResource.mRID (MBIE)	0..1 ResourceID_String	The unique identification of a resource. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a resource associated with a TimeSeries.

measurement_Unit.name (MBIE)	1..1	MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measurement used for the quantities expressed within the time series.
curveType (MBIE)	0..1	CurveType_String	The identification of the coded representation of the type of curve being described.

!# shows all association ends of TimeSeries with other classes.

Table !# - Association ends of Schedule assembly model::TimeSeries with other classes

mult.	mult.	Class type name	Description
Reason (MBIE)	[0..1]	Reason	At the TimeSeries level the reason code is used to enable processing of the reason text which, depending on market conditions, should be provided in intra day trading. In this context only one reason code has been defined (A48, modification reason). No other codes are permitted. Association Based On : Schedule contextual model:: TimeSeries.[] ----- Schedule contextual model:: Reason.Reason[0..1]

Global Used Datatypes

Datatypes

AreaID_String

The coded identification of a domain, i.e. balance area, grid area, etc.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for domain identification.

!# shows all attributes of AreaID_String.

Table !# - Attributes of ESMPDataTypes::AreaID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

CapacityContractKind_String

The coded identification of a contract type, e.g. daily auction, weekly auction, monthly auction, yearly auction, etc.

!# shows all attributes of CapacityContractKind_String.

Table !# - Attributes of ESMPDataTypes::CapacityContractKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ContractTypeList	Main Core value Space.

ID_String

A code to uniquely distinguish one occurrence of an entity from another.

In the ESMP context, the code is defined either by:

- an emitting company that provides an agreed identification unique within a business context such as capacity auction identification, market agreement identification, etc.

- a party (originator of the exchange) that provides a unique identification in the framework of a business exchange such as document identification, time series identification, bid identification, ...

!# shows all attributes of ID_String.

Table !# - Attributes of ESMPDataTypes::ID_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	String	Main Core value Space.

MeasurementPointID_String

The coded identification of a domain covering a number of related objects, such as metering point, accounting point, etc.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for measurement point identification.

!# shows all attributes of MeasurementPointID_String.

Table !# - Attributes of ESMPDataTypes::MeasurementPointID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

PartyID_String

The identification of an actor in the energy market.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for market participant identification.

!# shows all attributes of PartyID_String.

Table !# - Attributes of ESMPDataTypes::PartyID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

MarketRoleKind_String

The identification of the role played by a party.

!# shows all attributes of MarketRoleKind_String.

Table !# - Attributes of ESMPDataTypes::MarketRoleKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	RoleTypeList	Main Core value Space.

MeasurementUnitKind_String

The coded identification of a unit of measure that is applied to a quantity. The measurement units shall be in compliance with UN/ECE Recommendation 20.

!# shows all attributes of MeasurementUnitKind_String.

Table !# - Attributes of ESMPDataTypes::MeasurementUnitKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	UnitOfMeasureTypeList	Main Core value Space.

Position_Integer

An integer value, this value is used as a sequential value representing the relative position of an entity within a space such as a time interval.

!# shows all attributes of Position_Integer.

Table !# - Attributes of ESMPDataTypes::Position_Integer.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	Integer	Main Core value Space.

ProcessKind_String

The coded identification of the nature of process.

!# shows all attributes of ProcessKind_String.

Table !# - Attributes of ESMPDataTypes::ProcessKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ProcessTypeList	Main Core value Space.

ClassificationKind_String

The coded identification of the classification mechanism used to group a set of objects together. The grouping may be of a detailed or a summary nature.

!# shows all attributes of ClassificationKind_String.

Table !# - Attributes of ESMPDataTypes::ClassificationKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ClassificationTypeList	Main Core value Space.

ReasonCode_String

The coded motivation of an act.

!# shows all attributes of ReasonCode_String.

Table !# - Attributes of ESMPDataTypes::ReasonCode_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ReasonCodeTypeList	Main Core value Space.

ReasonText_String

The textual explanation of an act as a string of characters.

!# shows all attributes of ReasonText_String.

Table !# - Attributes of ESMPDataTypes::ReasonText_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype,IsBasedOn)	1..1	String	Main Core value Space.

ResourceID_String

The identification of a resource object in the energy market.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for resources (generator, lines, substations, etc.) identification.

!# shows all attributes of ResourceID_String.

Table !# - Attributes of ESMPDataTypes::ResourceID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

ESMPVersion_String

In ESMP, the coded value is restricted to digits.

A code that distinguishes one evolution of an identified object from another. Information about a specific object may be sent several times, each transmission being identified by a different version number.

!# shows all attributes of ESMPVersion_String.

Table !# - Attributes of ESMPDataTypes::ESMPVersion_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	String	Main Core value Space.

MessageKind_String

The coded type of a document.

!# shows all attributes of MessageKind_String.

Table !# - Attributes of ESMPDataTypes::MessageKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	MessageTypeList	Main Core value Space.

ESMP_DateTime

In ESMP, the dateTime shall be expressed in UTC as YYYY-MM-DDThh:mm:ssZ.

!# shows all attributes of ESMP_DateTime.

Table !# - Attributes of ESMPDataTypes::ESMP_DateTime.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DateTime	Main Core value Space.

YMDHM_DateTime

In ESMP, the date and time is expressed as "YYYY-MM-DDThh:mmZ", which conforms with ISO 8601 UTC time zone. This date and time is without the seconds.

!# shows all attributes of YMDHM_DateTime.

Table !# - Attributes of ESMPDataTypes::YMDHM_DateTime.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DateTime	The date and time as "YYYY-MM-DDThh:mmZ", which conforms with the ISO 8601 UTC time zone.

BusinessKind_String

The coded identification of the business type.

!# shows all attributes of BusinessKind_String.

Table !# - Attributes of ESMPDataTypes::BusinessKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	BusinessTypeList	Main Core value Space.

EnergyProductKind_String

The identification of the nature of an energy product such as power, energy, reactive power, etc.

!# shows all attributes of EnergyProductKind_String.

Table !# - Attributes of ESMPDataTypes::EnergyProductKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	EnergyProductTypeList	Main Core value Space.

ObjectAggregationKind_String

The coded identification of the aggregation object.

!# shows all attributes of ObjectAggregationKind_String.

Table !# - Attributes of ESMPDataTypes::ObjectAggregationKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ObjectAggregationTypeList	Main Core value Space.

CurveType_String

The coded identification of the type of curve.

!# shows all attributes of CurveType_String.

Table !# - Attributes of ESMPDataTypes::CurveType_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	CurveTypeList	Main Core value Space.

Enumerations

CodingSchemeTypeList

Codification scheme used to identify the coding scheme used for the set of coded values to identify specific objects.

Literal	Description	Code
EIC	The coding scheme is the Energy Identification Coding Scheme (EIC), maintained by ENTSO-E.	A01
Coding scheme which are part of GS1 system	The coding schemes maintained by GS1.	A10
Andorra National coding scheme	The National coding scheme of the country in question.	NAD
Albania National coding scheme	The National coding scheme of the country in question.	NAL
Armenia National coding scheme	The National coding scheme of the country in question.	NAM
Austria National coding scheme	The National coding scheme of the country in question.	NAT
Azerbaijan National coding scheme	The National coding scheme of the country in question.	NAZ
Bosnia and Herzegovina National coding scheme	The National coding scheme of the country in question.	NBA
Belgium National		

coding scheme	The National coding scheme of the country in question.	NBE
Bulgaria National coding scheme	The National coding scheme of the country in question.	NBG
Switzerland National coding scheme	The National coding scheme of the country in question.	NCH
Serbia and Montenegro National coding scheme	The National coding scheme of the country in question.	NCS
Czech Republic National coding scheme	The National coding scheme of the country in question.	NCZ
Germany National coding scheme	The National coding scheme of the country in question.	NDE
Denmark National coding scheme	The National coding scheme of the country in question.	NDK
Estonia National coding scheme	The National coding scheme of the country in question.	NEE
Spain National coding scheme	The National coding scheme of the country in question.	NES
Finland National coding scheme	The National coding scheme of the country in question.	NFI
France National coding scheme	The National coding scheme of the country in question.	NFR
United Kingdom National coding scheme	The National coding scheme of the country in question.	NGB
Georgia National coding scheme	The National coding scheme of the country in question.	NGE
Gibraltar National coding scheme	The National coding scheme of the country in question.	NGI
Greece National coding scheme	The National coding scheme of the country in question.	NGR
Croatia National coding scheme	The National coding scheme of the country in question.	NHR
Hungary National coding scheme	The National coding scheme of the country in question.	NHU
Ireland National coding scheme	The National coding scheme of the country in question.	NIE
Italy National coding scheme	The National coding scheme of the country in question.	NIT
Kyrgyzstan National coding scheme	The National coding scheme of the country in question.	NKG
Kazakhstan National coding scheme	The National coding scheme of the country in question.	NKZ
Liechtenstein National coding scheme	The National coding scheme of the country in question.	NLI
Lithuania National coding scheme	The National coding scheme of the country in question.	NLT
Luxembourg National coding scheme	The National coding scheme of the country in question.	NLU
Latvia National coding scheme	The National coding scheme of the country in question.	NLV
Morocco National coding scheme	The National coding scheme of the country in question.	NMA
Moldavia National coding scheme	The National coding scheme of the country in question.	NMD
Macedonia National coding scheme	The National coding scheme of the country in question.	NMK
Netherlands National coding scheme	The National coding scheme of the country in question.	NNL
Nordic Regional coding scheme	The coding scheme of the Nordic region which covers Denmark, Finland, Norway and Sweden.	NNN
Norway National coding scheme	The National coding scheme of the country in question.	NNO
Poland National coding scheme	The National coding scheme of the country in question.	NPL
Portugal National coding scheme	The National coding scheme of the country in question.	NPT
Romania National coding scheme	The National coding scheme of the country in question.	NRO
Russian Federation National coding scheme	The National coding scheme of the country in question.	NRU
Sweden National		

coding scheme	The National coding scheme of the country in question.	NSE
Slovenia National coding scheme	The National coding scheme of the country in question.	NSI
Slovakia National coding scheme	The National coding scheme of the country in question.	NSK
Turkey National coding scheme	The National coding scheme of the country in question.	NTR
Ukraine National coding scheme	The National coding scheme of the country in question.	NUA
CGM	The coding scheme used for Common Grid Model Exchange Standard (CGMES).	A02
Cyprus National coding scheme	The National coding scheme of the country in question.	NCY
ISO	The coding scheme for the preceding attribute is a code maintained by International Organization for Standardization (ISO), such as â€œISO 3166 *English country names and code elements*â€ .	A03

ContractTypeList

The contract type defines the conditions under which the capacity is allocated and handled, e.g. daily auction, weekly auction, monthly auction, yearly auction, etc.
The significance of this type is dependent on area specific coded working methods.

Literal	Description	Code
Daily contract	The condition under which capacity is allocated and handled is by daily auction or a daily transmission allocation procedure.	A01
Weekly contract	The condition under which capacity is allocated and handled is by weekly auction or a weekly transmission allocation procedure.	A02
Monthly contract	The condition under which capacity is allocated and handled is by monthly auction or a monthly transmission allocation procedure.	A03
Yearly contract	The condition under which capacity is allocated and handled is by yearly auction or a yearly transmission allocation procedure.	A04
Total contract	This is the sum of all capacity contract types for the period covered.	A05
Long term contract	The condition under which capacity is allocated and handled is by long term trade agreements according to European regulations (EU Directive 1228/2003).	A06
Intraday contract	The condition under which the capacity is allocated and handled is through an intraday auction and allocation process.	A07
Quarter yearly contract	The condition under which capacity is allocated and handled is by quarter yearly auction or a quarter yearly transmission allocation procedure.	A08
Semestrial contract	The condition under which capacity is allocated and handled is by half yearly auction or a half yearly transmission allocation procedure.	A09
Multiple year contract	The condition under which capacity is allocated and handled is by multiple year auctions.	A10
Intraday balancing mechanism contract	The condition under which the capacity is allocated and handled is through intraday energy balancing services.	A11
Historical contract	A Contract established before the EU directive that are valid until the term of the contract.	A12
Hourly contract	The condition under which capacity is allocated and handled is by hourly auctions.	A13
First intraday auction contract	The first intraday auction contract (IDA1).	A14
Second intraday auction contract	The second intraday auction contract (IDA2).	A15
Third intraday auction contract	The third intraday auction contract (IDA3).	A16

RoleTypeList

Identification of the role played by a party.

Literal	Description	Code
Trade responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A01
Consumption responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A02
Combined power exchange (not to	This role is no longer in the ENTSO-E Harmonised Role Model Document.	A03

be used)		
System operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A04
Imbalance settlement responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A05
Production responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A06
Transmission capacity allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A07
Balance responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A08
Metered data aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A09
Billing agent	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A10
Market operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A11
Energy supplier	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A12
Consumer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A13
Control area operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A14
Control block operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A15
Coordination centre operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A16
Grid access provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A17
Grid operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A18
Meter administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A19
Party connected to grid	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A20
Producer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A21
Profile maintenance party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A22
Meter operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A23
Metered data collector	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A24
Metered data responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A25
Metering point administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A26
Resource Provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A27
Scheduling coordinator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A28
Capacity Trader	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A29
Interconnection Trade Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A30
Nomination Validator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A31
Market information aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that collects information from different sources and assembles it to provide a summary of the market.	A32
Information receiver	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party, not necessarily a market participant, which receives information about the market.	A33
Reserve Allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.	A34
MOL Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that Informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.	A35
Capacity Coordinator	A party, acting on behalf of the SOs involved, responsible for establishing a coordinated Offered Capacity and/or NTC and/or ATC between several Scheduling Areas. Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A36

Reconciliation Accountable	A party that is financially accountable for the reconciled volume of energy products for a profiled local metering point.	A37
Reconciliation Responsible	A party that is responsible for reconciling, within a metering grid area, the volumes used in the imbalance settlement process for profiled metering points and the actual metered quantities.	A38
Data provider	A party that is responsible for providing information to a central authority.	A39
Local Issuing Office (LIO)	A party that is responsible for operating a Local Issuing Office (LIO).	A40
Central Issuing Office (CIO)	A party that is responsible for operating a Central Issuing Office (CIO).	A41
EIC Participant	A party that participates in the EIC environment.	A42
Weather analyser	A party that analyses the current and forecast weather situation and establishes a prognosis of its impact on the renewable energy environment as well as the overall load.	A43
Regional Security Coordinator (RSC)	The RSC as defined in the System Operation guideline.	A44
Energy Service Company (ESCO)	A party offering energy-related services to the Party Connected to Grid, but not directly active in the energy value chain or the physical infrastructure itself. The ESCO may provide insight services as well as energy management services.	A45
Balancing Service Provider	A party with reserve-providing units or reserve-providing groups able to provide balancing services to LFC Operators.	A46
Energy trader	A party that is selling or buying energy.	A47
LFC Operator	A party that is responsible for the Load Frequency Control of its LFC Area or LFC Block.	A48
Transmission System Operator (TSO)	The Transmission System Operator (TSO) is responsible for the transport of electricity on the extra high-voltage and high-voltage interconnected system. This is a market participant and not a role in Harmonised Role Model.	A49
Distribution System Operator (DSO)	Distribution System Operator (DSO) is responsible for transport of electricity on high-voltage (optionally), medium-voltage and low-voltage distribution systems. This is a market participant and not a role in Harmonised Role Model.	A50
Resource Capacity Mechanism Operator	Resource capacity mechanism operator is the party responsible to operate the resource capacity mechanism in a member state. It can either be the TSO or an independent party.	A51
Resource aggregator	A party that aggregates resources for usage by a service provider for energy market services.	A52
Cost sharing calculator	A party responsible for the mapping of the costs between areas and parties.	A53
Settlement responsible	A party responsible for settling the costs.	A54
Coordinated Capacity Calculator	Coordinated Capacity Calculator is the entity or entities with the task of calculating transmission capacity, at regional level or above.	A55
Flexibility service provider	A party that offers flexibility services based on acquired (aggregated) resources.	A56
Data access provider	A party responsible for facilitating access to data.	A57
Metered data administrator	A party responsible for storing and distributing validated measured data.	A58
Permission administrator	A party responsible for administrating a register of data access permissions for a set of metering points, making this information available to final customers and eligible parties in the sector, on request.	A59

UnitOfMeasureTypeList

(synonym MeasurementUnit) The unit of measure that is applied to a quantity. The measurement units shall be in compliance with UN/ECE Recommendation 20.

Literal	Description	Code
ampere	The unit of electrical current in the International system of Units (SI) equivalent to one Coulomb per second.	AMP
megavolt-ampere	MVA unit as per UN/CEFACT recommendation 20.	MVA
One	A unit for dimensionless quantities, also called quantities of dimension one.	C62
cubic hectometres	A unit of volume equal to one million cubic metres.	HMQ
kilovolt	kV unit as per UN/CEFACT recommendation 20.	KVT
kilometre	km unit as per UN/CEFACT recommendation 20.	KMT

kilovolt ampere reactive	A unit of electrical reactive power represented by a current of one thousand amperes flowing due to a potential difference of one thousand volts where the sine of the phase angle between them is 1. The unity power factor is expressed in thousands of a volt ampere reactive.	KVR
gigawatt hour	GWh unit as per UN/CEFACT recommendation 20.	GWH
kilowatt hour	A total amount of electrical energy transferred or consumed in one hour.	KWH
kilowatt	A unit of bulk power flow, which can be defined as the rate of energy transfer /consumption when a current of 1000 amperes flows due to a potential of 1000 volts at unity power factor expressed in thousands of a watt.	KWT
megavolt ampere reactive hours	Total amount of reactive power across a power system.	MAH
megavolt ampere reactive	A unit of electrical reactive power represented by a current of one thousand amperes flowing due to a potential difference of one thousand volts where the sine of the phase angle between them is 1.	MAR
megawatt	A unit of bulk power flow, which can be defined as the rate of energy transfer /consumption when a current of 1000 amperes flows due to a potential of 1000 volts at unity power factor expressed in millions of a watt.	MAW
cubic metres per second	The volume flow rate of cubic metre per second.	MQS
cubic metre	A Cubic metre.	MTQ
metre	The length of a metre.	MTR
megawatt hours	The total amount of bulk energy transferred or consumed.	MWH
percent	A unit of proportion equal to 0.01.	P1
gigawatt	GW unit as per UN/CEFACT recommendation 20.	A90
watt	The watt is the International System of Units (SI) standard unit of power (energy per unit time), the equivalent of one joule per second.	WTT
meter per second	A unit of measurement of the speed expressed in m/s.	MTS
degree (unit of angle)	A unit of measurement of angles expressed in a 0 to 360 degree gradient.	DD
watt per square meter	A unit of measurement of the density of heat flow rate expressed in watt per square meter.	D54
Celsius	A unit of measurement of temperature expressed in degree Celsius.	CEL
OKTA unit	A unit of measurement of the cloudiness expressed in OKTA or OCTA, i.e. A unit of count defining the number of eighth-parts as a measure of the celestial dome cloud coverage.	A59
hectopascal	A unit of measurement of the pressure expressed in hectopascal.	A97
millimeter	A unit of measurement of length expressed in millimeter.	MMT
K (Kelvin)	Temperature unit refer ISO 80000-5 (Quantities and units, Part 5: Thermodynamics).	KEL
Hertz	HTZ unit as per UN/CEFACT recommendation 20.	HTZ
Megawatt per Hertz	A unit of energy expressed as the load change in million watts that will cause a frequency shift of one hertz.	E08
millihertz	A unit of frequency equal to 0.001 cycle per second.	MTZ
minute	A period of time equal to sixty seconds.	MIN
second	A period of time equal to one second.	SEC

ProcessTypeList

Indicates the nature of process that the document addresses.

Literal	Description	Code
Day ahead	The information provided concerns a day ahead process.	A01
Intra day incremental	The information provided concerns an intra day schedule.	A02
Inter-area transit	The information provided concerns an inter area transit schedule. The rules governing this process are market dependent	A03
System operation closure	The information provided concerns the closure of a given period of both scheduled and regulation information.	A04
Metered data aggregation	The information provided concerns the aggregation process of metered information.	A05
	The information provided concerns the imbalance settlement for a given period for a	

Imbalance settlement	balance responsible party or parties.	A06
Capacity allocation	The information provided concerns the capacity allocation process.	A07
Central reconciliation	The process carried out to finalise the imbalance settlement based on actual metered values against provisional values from profiled metering points.	A08
Released capacity allocation	The process concerns the notification of capacity rights that are being released.	A09
Proposed capacity allocation	The process concerns the proposed capacity to be allocated for a given border.	A10
Agreed capacity allocation	The process concerns the capacity that has been agreed for allocation for a border.	A11
Long term	The process concerns scheduling all schedules except daily and intraday contracts.	A12
Post scheduling adjustment	The process concerns the adjustments made to previous schedules after the period of execution.	A13
Forecast	The data contained in the document are to be handled in short, medium, long term forecasting process.	A14
Capacity determination	The process of determining the capacity for use.	A15
Realised	The process for the treatment of realised data as opposed to forecast data.	A16
Schedule day	The process concerns the day ahead, intraday and eventually ex-post scheduling in a single document. The schedule will be transferred within the total position including historic information.	A17
Intraday total	This process concerns an intraday schedule which contains the accumulated day ahead and intraday current position.	A18
Intraday accumulated	This process concerns a single intraday schedule process where only intraday evolutions occur through version changes.	A19
SOMA process	System Operator meter alignment process.	A20
SOVM process	System Operator validated measurement process.	A21
RGCE accounting process	The document provides ENTSO-E Regional Group Continental Europe accounting process information.	A22
CCSR RGCE Settlement	The process concerns the control center settlement report for the whole of the ENTSO-E Regional Group Continental Europe.	A23
CBSR Settlement	The process concerns the control block settlement report.	A24
CASR Settlement	The process concerns the control area settlement report.	A25
Outage information	The process concerns TSO publication of outages on its power system.	A26
Reserve resource process	The process being described is for general reserve resources.	A27
Primary reserve process	The process being described is for primary reserves.	A28
Secondary reserve process	The process being described is for secondary reserves.	A29
Tertiary reserve process	The process being described is for tertiary reserves.	A30
Week ahead	The process being described is for the week ahead.	A31
Month ahead	The process being described is for the month ahead.	A32
Year ahead	The process being described is for the year ahead.	A33
Contracted	The process being described is for contracted information.	A34
Network information	The process being described is for network information.	A35
Creation	The process being described is for the creation of information.	A36
Modification	The process being described is for the modification of information.	A37
Deactivation process	The process being described is for deactivation of information.	A38
Synchronisation process	The process being described is for the synchronisation of information.	A39
Intraday process	The process being described is for intraday process.	A40
Redispatch process	The process being described is for redispatch activation.	A41
Activation history process	The process being described is for the provision of an activation history.	A42
Flow based domain constraint day-ahead	The information provided concerns the flow-based process in day ahead.	A43
Flow based domain constraint intraday	The information provided concerns the flow-based process in intraday.	A44
Two days ahead	Two days ahead.	A45
Replacement reserve	A process being described is for replacement reserves (RR) to restore or support the required level of frequency restoration reserves.	A46
Manual frequency restoration reserve	A process being described is for manual frequency restoration reserves (mFRR).	A47

Day-ahead capacity determination	The process run at the day-ahead timeframe to determine the capacity for use.	A48
Intraday capacity determination	The process run at the intraday timeframe to determine the capacity for use.	A49
Long term capacity determination	The process run at the long term timeframe to determine the capacity for use.	A50
Automatic frequency restoration reserve	A process being described is for automatic frequency restoration reserves (aFRR).	A51
Frequency containment reserve	A process being described is for frequency containment reserves (FCR).	A52
Common Grid Model (CGM) merging process	The process for merging individual grid models to form the common grid model.	A53
Coordinated security analysis	The process being described is used for coordinated security analysis.	A54
Exchange of master data	A process for exchanging master data.	A55
Frequency restoration reserve	The process being described is for general frequency restoration reserve.	A56
FSKAR settlement	The information provided concerns the Financial Settlement of K?f, ACE and ramping period settlement for a given period.	A57
mFRR capacity market	Processes related to the Reserve option market to assure that there are enough available reserves for the manual FRR market.	A58
Internal trade reporting	The process related to internal trade reporting.	A59
Scheduled activation mFRR	mFRR being subject to scheduled activation.	A60
Direct activation mFRR	mFRR being subject to direct activation.	A61
Registration	A process related to the registration and management of object information.	A62
Imbalance Netting	The process described is for imbalance netting.	A63
Criteria application for instantaneous frequency	The process describes criteria application for instantaneous frequency.	A64
Criteria application for frequency restoration	The process describes criteria application for frequency restoration.	A65
Cost sharing	The process describes the cost sharing process for costly remedial actions.	A66
Central selection aFRR	aFRR subject to central selection of bids for activation.	A67
Local selection aFRR	aFRR subject to local selection of bids for activation.	A68
Common grid model alignment	The process for aligning bidding zone net positions.	A69
Pan-European verification function	The process for verifying bidding zone net positions.	A70
Outage planning coordination	The process being described is used for outage planning coordination.	A71
Short term adequacy	The process being described is used for short term adequacy calculations.	A72
Coordinated capacity calculation	The process being described is used for coordinated capacity calculation.	A73
Access to metered data	Access to meter readings of electricity consumption from or electricity feed into the grid.	A74
Permission administration	Administration of permissions to access and/or use data.	A75

ClassificationTypeList

Indicates the classification mechanism used to group a set of objects together. The grouping may be of a detailed or a summary nature.

Literal	Description	Code
Detail type	The Time Series content provides detailed information.	A01
Summary type	The Time Series content provides aggregated information.	A02

ReasonCodeTypeList

The coded motivation of an act.

Literal	Description	Code
Errors not specifically identified	This code is used to identify errors that have not been specifically addressed in the Reason code list. It can be used at any level and refers to the level for which it has been identified.	999

Message fully accepted	The message has been fully accepted for application processing.	A01
Message fully rejected	No part of the message has been accepted for application processing, e.g. Global position incomplete.	A02
Message contains errors at the time series level	Part of the message contents, i.e. certain time series, has been accepted for application processing. It is necessary to look at the time series level to determine the time series that have been rejected. The time series is excluded from the global position.	A03
Time interval incorrect	The schedule time interval is not within the contractual agreement or the period does not agree with the schedule time interval.	A04
Sender without valid contract	The sender has no current valid contract with the TSO. The message consequently will be fully rejected.	A05
Schedule accepted	The schedule of the recipient as presented has been completely accepted.	A06
Schedule partially accepted	The schedule of the recipient as presented has been partially accepted. It is necessary to look at the time series level to determine the changes (time series rejected, modified, etc.).	A07
Schedule rejected	The schedule of the recipient as presented has been totally rejected. The cause could be the non presentation of a counter part for the involved trades.	A08
Time series not matching	Time series mismatches.	A09
Credit limit exceeded	The contractual credit limit has been exceeded and as such the message has been rejected.	A10
Time series fully rejected	The time series has been fully rejected. In the case of a confirmation report, this reason code is used in conjunction with either A26 or A30.	A20
Time series accepted with specific time interval errors	The time series has been accepted but some time interval quantities have been rectified or zeroed out.	A21
In party/Out party invalid	There is no contract for the parties indicated or the rules for cross border nominations are not being respected. The time series has been rejected.	A22
Area invalid	The area is unknown or not allowed. The time series has been rejected.	A23
A24 not applicable	This code is no longer applicable.	A24
A25 not applicable	This code is no longer applicable.	A25
Default time series applied	The time series has been rejected and replaced with a default time series profile. This reason code may not be used in conjunction with A30.	A26
Cross border capacity exceeded	The cross border capacity has been exceeded. The time series has been rejected or adjusted.	A27
Counterpart time series missing	This provides an indication that the time series has not got a counterpart time series. In the case of an Intermediate Confirmation Report this is advising the recipient that the time series may be rejected at nomination closure if the counterpart time series is not received. In the case of a Final Confirmation Report this is informing the recipient that the time series has been rejected because the counterpart time series has not been forthcoming.	A28
Counterpart time series quantity differences	The time series has been rejected as it does not match that of the counterpart who is considered by market rules to be correct.	A29
Imposed Time series from nominated party's time series (party identified in reason text)	The nominated party's time series has replaced the current time series. This reason code may not be used in conjunction with A26.	A30
Resolution inconsistency	The resolution is not coherent with the time interval, or resolution not valid.	A41
Quantity inconsistency	The quantity is not coherent. For example a time period with the same version number but different quantities or an non permitted number of digits after the decimal point, etc.	A42
Quantity increased	The quantity has been increased in order to satisfy minimum constraints.	A43
Quantity decreased	The quantity has been decreased in order to satisfy congestion constraints.	A44
Default quantity applied	The default quantity has been applied as the current quantity does not satisfy contractual obligations.	A45
Quantities must not be signed values	The quantity proposed is illegal since signed values are only permitted in specific circumstances.	A46
A47 not applicable	This code is no longer applicable.	A47
Modification reason	In an intraday transmission, the reason for the modification is as follows (in the reason text).	A48

Position inconsistency	A position is missing or too many.	A49
Senders time series version conflict	There is an error in the sender time series version, i.e. it could be superior to the message version or is inconsistent with the existing data. The time series has been rejected.	A50
Message identification or version conflict	The message identification is already in the receiving system. Or a higher version already exists. Message rejected.	A51
Time series missing from new version of message	A time series is not contained in a new version of the message. Message rejected.	A52
Receiving party incorrect	The receiving party is incorrect. Message rejected.	A53
Global position not in balance	The message does not balance out to zero. Market rules might require that the message is rejected.	A54
Time series identification conflict	The identification of the time series is duplicated or incorrect. Time series will be rejected.	A55
Corresponding Time series not netted	All corresponding time series must be netted. Time series rejected.	A56
Deadline limit exceeded/Gate not open	The deadline for receiving schedule messages has passed. Message or time series rejected.	A57
One to one nomination inconsistency	There is a one to one nomination inconstancy with the in/out parties or areas. Time series rejected.	A58
Not compliant to local market rules	The level in which this is identified is not in compliance with local market rules. The level in question has been rejected.	A59
Inter-area transit schedule exceeds nominated schedule	The inter-area transit schedule exceeds the nominated schedule for the same time interval. The inter-area transit schedule is rejected.	A60
Currency invalid	The currency is not in compliance with ISO 4217.	A61
Invalid business type	The business type does not belong to the valid set of business types for the process in question.	A62
Time Series modified	The time series has been modified.	A63
Resource Object Invalid	The Resource Object defined in the document is not valid.	A64
Reserve object Technical limits exceeded	Reserve objects aggregated values are not within technical/prequalified limits	A65
Planned reserves do not correspond with contractual data	Planned reserves do not correspond with contractual data.	A66
Limit Data is not available	Limit Data is not available.	A67
Reserve Object not qualified for reserve type	Reserve Object is not qualified for reserve type.	A68
Mandatory attributes missing	Mandatory attributes missing.	A69
Curtailment	The capacity in question has been curtailed.	A70
Linked bid rejected due to associated bid unsuccessful	The bid in question has been rejected because an associated bid has been unsuccessful.	A71
Original bid divided to permit acceptance	The original bid quantity has been divided to enable it to be accepted.	A72
Bid accepted	The bid in question has been accepted.	A73
Auction Status	The information in the Reason Text provides auction status information.	A74
Right status information	The information in the Reason Text provides status information concerning the transmission rights in question.	A75
Agreement identification inconsistency	There is an inconsistency between the contract type and the agreement identification.	A76
Dependency matrix not respected	There is an inconsistency between the document contents and the dependency matrix.	A77
Sender identification and/or role invalid	The identification of the sender or the sender/role combination is invalid.	A78
Process type invalid	The process type does not figure in the list of valid process types for this document.	A79
Domain invalid	The domain does not figure in the list of valid domains for this document and process.	A80
Matching period invalid	The period is not within the expected limits.	A81
In/Out area inconsistent with domain	The in and out area does not figure within the domain specified.	A82
Disagree with matching		

results	The matching results provided are not consistent.	A83
Confirmation ignored due to higher version already received	The report has been ignored since a higher version has been received.	A84
Confirmation without adjustment (time series have been matched without change)	The report has been successfully matched without any changes.	A85
Confirmation with adjustment (time series have been modified)	The report has been matched but required adjustment.	A86
For action (only in intermediate confirmation - time series need mutual agreement and action)	The report in question is only for action in an intermediate stage.	A87
Time series matched	The time series has been successfully matched.	A88
Time series ignored (note: this can only apply to time series that are set to zero - see matching principles)	The time series has been ignored and not matched since it does not figure in a counterparty transmission. All are correctly equal to zero.	A89
Modification proposal (intermediate confirmation)	The document is a proposal for change before finalization.	A90
Expected document not received	The document that is expected has not been received within the expected timeframe.	A91
Not possible to send document on time, but estimated delivery time is provided	The document that is due cannot be transmitted within the required timeframe. An estimated time of transmission is provided.	A92
Not possible to send document on time, and furthermore no expected time of return to normal situation	The document that is due cannot be transmitted within the required timeframe. The time of transmission of the document is unknown.	A93
Document cannot be processed by receiving system	The receiving system cannot process that document in question.	A94
Complementary information	Additional text is provided in order to further explain a condition, for example to provide details of an outage.	A95
Technical constraint	A technical constraint has been applied.	A96
Force majeure curtailment	Curtailment due to Force Majeure. A code that enables the identification of the curtailment reason for settlement purposes.	A97
Network security curtailment	Curtailment due to network security reasons A code that enables the identification of the curtailment reason for settlement purposes.	A98
Auction cancelled	The auction has been cancelled.	A99
Incomplete document	The document is incomplete and cannot be processed.	B01
Accounting Point (tie-line) Time Series missing	The document is incomplete as a time series for an accounting point is missing.	B02
Meter data Time series missing	The document is incomplete as a time series for meter data is missing.	B03
Estimated values not allowed in first transmission	The document is in its initial form and estimated values are not allowed.	B04
No quantity values allowed for a quality that is not available	No quantity values are allowed for a quality that is not available.	B05
Time series accepted	Time series accepted.	B06
Auction without bids being entered	The auction has terminated without any bids being submitted. The ReasonText may provide the identification of the auction in question.	B07
Data not yet available	It is not possible to perform the necessary action since the required data for this action is not yet available.	B08
Bid not accepted	The bid in question has not been accepted.	B09
Initiator area problem	The problem concerns the initiator area.	B10
Cooperating area problem	The problem concerns the cooperating area.	B11
Communication status currently active	The status within the system indicates that the communication capability is currently active.	B12

Communication status currently inactive	The status within the system indicates that the communication capability is currently inactive.	B13
Communication status currently restricted	The status within the system indicates that the communication capability is currently restricted.	B14
Problem associated with both areas	The problem concerns both areas.	B15
Tender unavailable in MOL list	A tender that has been requested is no longer available in the MOL.	B16
Price based on preliminary exchange rate	The price is based on a preliminary exchange rate.	B17
Failure	A failure has occurred.	B18
Foreseen maintenance	Maintenance is foreseen.	B19
Shutdown	A shutdown has occurred.	B20
Official exchange rate approved	The official exchange rate has been approved.	B21
System regulation	The information provided regards a regulation for system purposes.	B22
Frequency regulation	The information provided regards a regulation for frequency purposes.	B23
Load flow overload	Situation in the grid, where loading of a certain grid element, e.g. overhead line, is above defined technical limits.	B24
Voltage level adjustment	A TSO activity to maintain an acceptable voltage profile throughout the network. This is achieved by balancing of the respective reactive power requirements of the network and the customers.	B25
Emergency situation curtailment	Curtailment due to emergency situation. A code that enables the identification of the curtailment reason for settlement purposes.	B26
Calculation process failed	The calculation has not been performed.	B27
No capacity constraint impact on the market	The market position is such as no capacity constraint is applied to limit the cross border exchanges.	B28
Special Condition	Special condition need to be fulfilled.	B29
Unverified	Missing or not validated data.	B30
Verified	Data has successfully passed the verification process.	B31
CGM inconsistency	Describes an element which was not found in the CGM.	B32
Network dictionary inconsistency	Describes an element which was not found in the network dictionary.	B33
Capacity reduced by TSO	Describes a capacity that was reduced by a TSO.	B34
Overload	Describes an N-k or N state overload.	B35
GLSK limitation	Describes a situation in which there is not enough power adjustment in the GLSK file to cover the capacity.	B36
Voltage constraint	Describes an N-k or N state voltage violation.	B37
Angle constraint	Describes an N-k or N state angle violation.	B38
Stability	Describes a situation in which the dynamic behaviour of the network violated.	B39
Loadflow divergence	Describes a network situation in which the provided capacity values are part of a load flow divergence situation.	B40
Exclusion for SoS reasons	This is the adjustment applied to the capacity of a branch to have a minimum RAM (Remaining Available Margin) available for commercial exchanges.	B41
Constraint by the market	A constraint due to market restrictions.	B42
Ordinary	The contingency is ordinary (methodology for coordinating operational security analysis, article 6).	B43
Exceptional	The contingency is exceptional (methodology for coordinating operational security analysis, article 6).	B44
Out of range	The contingency is out of range (methodology for coordinating operational security analysis, article 6).	B45
Internal congestion	A temporary congestion within a bidding zone or scheduling area.	B46
Operational security constraints	Operational security constraints identified by TSOs.	B47
Estimated value	Describes a situation where a calculation process has failed and extrapolated or interpolated values will be applied.	B48
Balancing	Activated for balancing purposes.	B49
Values shared	Values of this time series are also valid for counter-party.	B50
Outside price limits	The offered price is not within the valid limits.	B51
Previous timeframe data	In case of processing issue, sent data are based on the previous timeframe.	B52

MOL merging succesful	The merging of the Merit Order List has been successfully processed.	B53
MOL merging failed	The merging of the Merit Order List has failed.	B54
Because of redispatching	Because of redispatching according to Article 2(26) of Commission Regulation (EU) 543/2013	B55
Because of countertrading	Because of countertrading according to Article 2(13) of Commission Regulation (EU) 543/2013	B56
Because of other remedial action	Not available because of any remedial action.	B57
Insufficiency of reserves	The reserve capacity is not sufficient.	B58
Unavailability of reserve providing unit	The unit providing the reserve is subject to technical unavailability.	B59
Unavailability of automatic protection systems	Unavailability of tools to detect predetermined system conditions that have a high probability of causing unusual stress on the power system, for which pre-planned remedial action is considered necessary or for which automatic protective actions may be triggered such as special protection schemes or automatic load shedding.	B60
Physical cable or converter restrictions	Limitation due to physical cable or converter restrictions.	B61
Constraints in controller systems	Limitation due to constraints in controller systems.	B62
Adjusted because of expected violation of operational security	The capacity is adjusted because of an expected violation of operational security limits of physical transmission assets.	B63
Adjusted because already considered remedial actions are assessed as not sufficient	The capacity is adjusted because the remedial actions were assessed as not sufficient or ineffective to avoid the expected violation of operational security limit(s).	B64
Time series partially accepted	The time series is partially accepted.	B65
Demand fully netted	Demand was fully netted against other demand(s).	B66
Bid activated in same direction	One or several bids were activated in the same direction within the same uncongested area as the demand.	B67
Optimization in progress	The activation optimisation framework is being executed.	B68
Wrong energy market timeframe	The given energy market timeframe is invalid or wrong.	B69
Message partially accepted	The detailed transactions of the received document are partially accepted. It is necessary to look at the detailed (transaction) level to determine if the transaction is accepted, rejected etc.	B70
(Explicit) Permission from the customer	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: The data subject has given consent to the processing of his or her personal data for one or more specific purposes.	B71
In execution of a contract with the customer (Execution of contract)	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for the performance of a contract to which the data subject is party or in order to take steps at the request of the data subject prior to entering into a contract.	B72
Legal obligations	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for compliance with a legal obligation to which the controller is subject.	B73
In protection of vital interests of the customer	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary in order to protect the vital interests of the data subject or of another natural person.	B74
In order to perform a duty of common (national) interest	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for the performance of a task carried out in the public interest or in the exercise of official authority vested in the controller.	B75
Justified interest of the dataprocessee	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for the purposes of the legitimate interests pursued by the controller or by a third party, except where such interests are overridden by the interests or fundamental rights and freedoms of the data subject which require protection of personal data, in particular where the data subject is a child	B76
Fulfilment of purpose	The purpose of the object or action has been fulfilled.	B77
Reach of end timestamp	The object or action has reached its end timestamp.	B78
Revocation	Revocation of the action or object.	B79
Termination	Termination of the action or object.	B80
No valid grid access	There is no valid grid access contract for a specific metering point. There is no valid	

contract	grid access contract for a specific metering point.	B81
Customer move-out	A customer has moved out.	B82

MessageTypeList

The coded type of a document. The message type describes the principal characteristic of a document.
This enumeration is used in the XML instances based on IEC 62325.

Literal	Description	Code
Balance responsible schedule	A schedule which has been prepared by a balance responsible party providing planned schedule information.	A01
Allocated capacity schedule	A schedule which has been prepared by a capacity allocator providing allocated capacity.	A02
Balance area schedule	A schedule that provides the planned schedule information for a balance area.	A03
System Operator area schedule	A compilation of all external schedules concerning two System Operator areas or a connector concerning two System Operator of all balance responsible parties.	A04
Control block area schedule	A compilation of all the exchange programs of all control areas for one control block with all neighbouring control areas of a neighbouring control block.	A05
Coordination center area schedule	A compilation of the exchange programs of all exchange blocks divided into UCTE south and north.	A06
Intermediate confirmation report	An intermediate confirmation report that may be produced between final cutoffs.	A07
Final confirmation report	A final confirmation report that is produced after a final cutoff.	A08
Finalised schedule	A compilation of a set of schedules that have been finalized after a given cutoff.	A09
Regulation data report	A compilation of the time series employed on a given day to ensure the balance of the system.	A10
Aggregated energy data report	A compilation of the time series of all the meter readings or their equivalent for a given period.	A11
Imbalance report	The report containing the complete situation of a given period for a party and including the schedules, regulation data and actual or calculated readings.	A12
Interconnection Capacity	Document for cross-border capacity exchanges.	A13
Resource Provider Resource Schedule	A document providing the schedules for resource objects submitted by a resource provider.	A14
Acquiring System Operator Reserve Schedule	A document providing reserve purchases submitted by an Acquiring System Operator.	A15
Anomaly Report	A document providing anomaly information for the receiving party to correct.	A16
Acknowledgement Document	A document providing acknowledgement information.	A17
Confirmation report	A document providing confirmation information.	A18
Capacity for Resale	A document providing information about capacity for resale.	A19
Approved Capacity Transfer	A document to approve a capacity transfer.	A20
Capacity transfer notification	A document notifying a capacity transfer.	A21
Transmission rights portfolio	A document providing the portfolio of the transmission capacity rights of a market participant.	A22
Allocations	A document providing the capacity allocations for a border.	A23
Bid document	A document providing bid information.	A24
Allocation result document	A document providing the allocation results of an auction.	A25
Capacity document	A document providing capacity information.	A26
Rights document	A document providing transmission capacity rights information.	A27
Generation availability schedule	This document contains information related to energy availability.	A28
Cross border schedule	This document contains the cross border schedules for all the borders of a given country where energy is exchanged.	A30
Agreed capacity	The capacity agreed between parties.	A31
Proposed capacity	The capacity proposed for agreement between parties.	A32
System vertical load	The sum of all flows out of the transmission grid via directly connected transformers and lines to distribution grids and end consumers as known by the System Operator.	A33
Escalation document		

	A document which requesting the escalation of a situation.	A34
Trouble shooting document	A document providing trouble shooting information for the resolution of a problem.	A35
Deactivation document	A document providing deactivation information.	A36
Reserve tender document	The document that is used for the tendering for reserves within the ERRP process.	A37
Reserve Allocation Result Document	The document used to provide the results of a Reserve auction.	A38
SATCR activation	The document is used to provide the activation of reserves through the SATCR process.	A39
DATCR activation	The document is used to provide the activation of reserves through the DATCR process.	A40
Activation response	The document is used to provide a response to a request to activate reserves.	A41
Tender reduction	The document is used to provide information concerning the reduction of a previously submitted tender.	A42
MOL Document	The document is used to provide Merit Order List information.	A43
Price Document	The document is used to provide market price information.	A44
Measurement Value Document	The document is used to provide measurement information from measurement devices.	A45
SOAM Document	The document is used to provide system operator accounting data for matching.	A46
SOVA Document	The document is used to provide system operator validated accounting data.	A47
CCVA Document	The document is used to provide coordination center validated accounting data.	A48
Daily settlement document	The document is used to provide daily settlement information.	A49
Weekly settlement document	The document is used to provide weekly settlement information.	A50
Capacity Auction Specification Document	The document is used to provide capacity auction specification information.	A51
Market Coupling Results Document	The document is used to provide the results of a market coupling auction.	A52
Outage publication Document	The document is used to provide the outage information for publication.	A53
Forced generation outage Document	A document providing information on forced generation outages.	A54
Summarised Market Schedule	A compilation of all external schedules concerning two Market Balance Areas of all balance responsible parties.	A55
Compensation Program Schedule	A schedule that provides the schedule information for the compensation of unintended deviation.	A56
Load Frequency Control Program Schedule	A schedule that provides the schedule information for the Load Frequency Control Program of a Control Area or a Control Block.	A57
Timeframe Independent Schedule	A compilation of all external Timeframe Independent Schedules concerning two System Operators.	A58
Information request	An information request being made concerning some specific information.	A59
status request for a position independently from a specific process	A status request concerning the position of an object independently of any ongoing processes.	A60
Estimated Net Transfer Capacity	The estimated net transfer capacity for a given border.	A61
Compensation rights	The capacity rights that have been allocated as compensation.	A62
Redispatch notice	A notice to confirm the actions agreed between System Operators to resolve a congestion situation through redispatch.	A63
Tender reduction response	A response to a tender reduction request that provides corrections to the initial document.	A64
System total load	Total load', including losses without power used for energy storage, is equal to generation deducted with exports, added with imports and deducted with power used for energy storage.	A65
Final MOL	A document providing the information concerning the situation of the MOL at the end of an activation period.	A66
Resource Provider Schedule for production/consumption	A document providing the schedules for production/consumption for resource objects submitted by a resource provider.	A67
Installed generation per type	A document providing the installed generation per generation type.	A68
Wind and solar forecast	A document providing the forecast of wind and solar generation.	A69
Load forecast margin	A document providing the load forecast margin for a period.	A70
Generation forecast	A document providing the generation forecast for a period.	A71
Reservoir filling		

information	A document providing information concerning the filling of reservoirs.	A72
Actual generation	A document providing the actual generation for a period.	A73
Wind and solar generation	A document providing the generation of wind and solar energy for a period.	A74
Actual generation per type	A document providing the actual generation per generation type for a period.	A75
Load unavailability	A document providing the unavailability of units providing load on the network.	A76
Production unavailability	A document providing the unavailability of production units providing energy to the network.	A77
Transmission unavailability	A document providing the unavailability of transmission capacity on the network.	A78
Offshore grid infrastructure unavailability	A document providing the unavailability of an offshore grid infrastructure to the network.	A79
Generation unavailability	A document providing the unavailability of generation units providing energy to the network.	A80
Contracted reserves	A document providing the reserves contracted for a period.	A81
Accepted offers	A document providing the offers of reserves that have been accepted for a period.	A82
Activated balancing quantities	A document providing the quantities of reserves that have been activated for balancing.	A83
Activated balancing prices	A document providing the prices of the reserves that have been activated for balancing.	A84
Imbalance prices	A document providing the prices of reserves due to imbalance for a period.	A85
Imbalance volume	A document providing the volume of the imbalance for a period.	A86
Financial situation	A document providing the financial situation for reserves.	A87
Cross border balancing	A document providing the cross border balancing requirements for a period.	A88
Contracted reserve prices	A document providing the price of reserves contracted for a period.	A89
Interconnection network expansion	A document providing information on the expansion of the interconnection network.	A90
Counter trade notice	A document providing information on counter trades for a period.	A91
Congestion costs	A document providing the cost of congestion for a period.	A92
DC link capacity	A document providing the DC links for a period.	A93
Non EU allocations	A document providing allocations made to non EU member states.	A94
Configuration document	A document providing configuration information.	A95
Redispatch activation document	A document enabling the activation of a redispatch notice.	A96
Detailed activation history document	A document enabling a detailed history of activations.	A97
Aggregated activation history document	A document enabling an aggregated history of activations.	A98
HVDC Link constraints	A document providing the information concerning the maximum and minimum active power flow through each link can limited.	A99
HVDC Configuration	A document providing the information concerning the power set point. The mode in which the HVDC is managed.	B01
HVDC Schedule	A document providing the information for operating DC links.	B02
EIC code request	A document providing the information requesting a new EIC code.	B03
EIC code information	A document providing EIC information in a central registry exchange or information to an EIC participant.	B04
EIC code publication	A document providing EIC publication information in a web site publication of a limited set of information.	B05
Critical network element determination	A document providing all the elements necessary for the capacity coordinator to determine the transfer capacity and the associated critical network elements.	B06
Critical network element publication	A document providing all the elements necessary for the market information aggregator and TSOs to know the critical network elements which limit the transfer capacity.	B07
Flow based domain	A document providing the capacity domain and its limits available for the TSO.	B08
Flow based domain publication	A document providing the capacity domain and its limits available for the market.	B09
Flow based domain market impact publication	A document providing the capacity domain and its impacts on the market to be published.	B10
Anonymized flow based parameters publication	A document providing all the relevant flow based parameters in case of flow based capacity allocation.	B11
Critical network element market impact publication	A document providing the critical network elements and its impacts on the market to be published.	B12

Weather document	An estimation or prediction in advance of the weather by analysis of meteorological data and the results of what actually happened with the weather.	B13
Energy prognosis document	A document to provide the prognosis of energy production/load for a given period.	B14
Network constraint document	A document providing the network constraint situations used for the load flow studies. A network constraint situation includes contingencies, monitored elements and remedial actions.	B15
Aggregated netted external market schedule document	A document used to report aggregated netted external market schedules for a given border.	B16
Aggregated netted external TSO schedule document	A document used to report aggregated netted external TSO schedules for a given border.	B17
Reporting status market document	A document used to report the status of aggregated netted external market schedules, aggregated netted external TSO schedules and compensation program schedules on a given border.	B18
Reporting information market document	A document used to report the information concerning aggregated netted external schedules, aggregated netted external market schedules, aggregated netted external TSO schedules, compensation program schedules, netted area position schedules and netted area AC position schedules to an interested party.	B19
Status request for a reporting information market document	A document requesting the provision of a reporting information document.	B20
Reserve need document	Used by a TSO to send its reserve needs.	B21
Generation and load shift keys document	A document providing the values of the generation and load shift keys to be used on network model.	B22
Offers to be activated	A document containing the outcome of the process, with the list of offers that are to be activated by the TSO concerned and the results for its reserve needs.	B23
Clearing price	A document containing the outcome of the process, with the clearing prices for a period.	B24
Security analysis report	A document providing a report on a performed security analysis.	B25
Aggregated netted external schedule document	A document used to report aggregated netted external schedules for a given border.	B26
External TSO schedule	A document used to report external TSO schedules for a given border or interconnector.	B27
Move of scheduled production	A document indication a move of scheduled production.	B28
PS&LC results document	A document providing Pole Splitting & Loss Calculation results.	B29
Notification data market document	A document used to notify data to any information receiver.	B30
Additional Constraint document	A document describing additional constraints for a capacity calculation process.	B31
Operational state document	A document used for exchanging operational states for grid assets.	B32
Published offered capacity	A document providing the most recent values of offered capacity.	B33
Market result document	Published prices and volumes	B34
Area Configuration document	A document containing the definition of areas.	B35
Area Composition document	A document containing the relations between areas, i.e which Metering Grid Areas a Bidding Zone composed of.	B36
Connected Areas document	A document containing which other areas an area is connected to i.e. which Metering Grid Areas a Metering Grid Area is connected to.	B37
Settlement document	A document providing settlement information.	B38
Imbalance prognosis document	A document to provide the prognosis of energy imbalances for a given period.	B39
Complete set of bids	Submission of complete set of bids. If there are existing bids, they should be replaced.	B40
Merged MOL notice	A notice providing information on the MOL merging process.	B41
K-factor document	A document providing K-factor values.	B42
Settlement coordination document	A document providing settlement information for coordination between different parties.	B43
Financial settlement document	A document providing financial settlement information.	B44
Bid availability document	A document providing the detailed reasons for changing the availability or volume of a bid.	B45
Resource capacity unit document	A document providing information about resource capacity units.	B46
Other market		

information	A document providing other market information.	B47
Permission document	A document that contains detailed information about permissions.	B48
Requested capacity	A document providing requested capacity (e.g. ATC, CZCL) value.	B49
Voltage control activation document	A document providing the number of generating units to activate for voltage control, by means of their operation in synchronous condenser mode.	B50
Transmission inventory document	A document providing a transmission inventory.	B51
Transmission inventory	A document providing statistics on the transmission grid.	B52

BusinessTypeList

The exact business nature identifying the principal characteristic of a time series.

Literal	Description	Code
Production	The nature of the business being described is production details.	A01
Internal trade	The nature of the business being described is internal trade details.	A02
External trade explicit capacity	The nature of the business being described is external trade details between two areas with limited capacity requiring a capacity agreement identification.	A03
Consumption	The nature of the business being described is consumption details.	A04
External trade total	The nature of the business being described is external trade total.	A05
External trade without explicit capacity	The nature of the business being described is external trade details between two areas without requiring capacity allocation information.	A06
Net Production / Consumption	Net production/consumption - where signed values will be used. With the following rules: In area=Out area, In party=Out party, + means production and - means consumption.	A07
Net internal trade	Net internal trade - where the direction from out party (seller) to in party (buyer) is positive and the opposite direction is negative (with minus signs).	A08
IPP (Independent Power Producer)	A time series concerning the production schedule from an IPP.	A09
Tertiary control	A time series concerning tertiary reserve.	A10
Primary control	A time series concerning primary reserve.	A11
Secondary control	A time series concerning secondary reserve.	A12
Load profile	A time series concerning a load profile as calculated by a metered data aggregator.	A13
Aggregated energy data	A time series concerning adjusted metered readings received from a metered data collector and aggregated and validated by a metered data aggregator.	A14
Losses	A time series concerning losses that have been calculated for a tieline or an area.	A15
Transits (CBT)	A time series concerning inter area transit flows determined for CBT requirements.	A16
Settlement deviation	A time series concerning the imbalance energy calculated by an imbalance settlement responsible.	A17
Technical constraint deviation	A time series defining the imbalance between schedules accepted by the system operator due to technical constraints and schedules declared by the balance responsible party.	A18
Balance energy deviation	A time series defining the imbalance between the schedule of a balance responsible party that has been corrected by the system operator after using balance energy bids and the schedule that was accepted by the system operator due to technical constraints.	A19
Imbalance volume	A time series defining the imbalance between the actual meter readings and the schedule of the balance responsible party corrected by the system operator after using balance energy bids.	A20
Unintended energy	A timeseries concerning the volume of an unintended cross-border exchange of energy.	A21
Frequency control	A time series concerning primary and secondary reserve.	A22
Balance management	A time series concerning energy balancing services.	A23
Total trade	A time series concerning the total of both the internal and external trades.	A24
General Capacity Information	A time series providing the total capacity available on a TSO border.	A25
Available transfer capacity (ATC)	Available transfer capacity for cross-border exchanges.	A26
Net transfer capacity (NTC)	Net transfer capacity for cross-border exchanges.	A27
Control Area Program	A time series providing the total exchanges between two TSOs (including the commercial transactions, the compensation program and the losses compensation program). Note this definition might change when UCTE brings forward its coding requirements.	A28
Already allocated		

capacity (AAC)	The already allocated capacity is the total amount of allocated transmission rights.	A29
Internal inter area trade	A trade that occurs between internal areas within a market balance area.	A30
Offered Capacity	The time series provides the offered capacity.	A31
Capacity transfer notification	The time series provides information concerning the notification of the transfer of capacity to another market participant.	A32
Authorised AAC	The time series in question provides the amount of transmission capacity rights to be nominated.	A33
Capacity rights	The time series in question provides the capacity rights allocated for a given border.	A34
Minimum authorised AAC	The time series in question provides the minimum amount of transmission capacity rights to be nominated.	A35
Maximum authorised AAC	The time series in question provides the maximum amount of transmission capacity rights to be nominated.	A36
Installed generation	The time series in question provides the installed generation.	A37
Available generation	The time series in question provides the available generation.	A38
Interconnection Trade Responsible Designation	The Time series in question provides the designation of the ITR that may nominate the capacity in question.	A40
Released AAC	The already allocated capacity (AAC) that has been released for resale.	A41
Requested capacity (with price)	The time series in question provides information concerning the requested capacity including price information.	A42
Requested capacity (without price)	The time series in question provides information concerning the requested capacity but excludes price information.	A43
Compensation program	Compensation of unintentional deviation is performed by exporting to / importing from the interconnected system during the compensation period by means of schedules as calculated during the accounting of unintentional deviations.	A44
Schedule activated reserves	The cross border or internal reserves that are to be activated through schedule nomination.	A45
System Operator redispatching	The cross border redispatching between System Operators that are to be activated through schedule nomination.	A46
Market capacity price	The price of the capacity offered on a given market.	A47
Market capacity price differential	The difference between the price of capacity in a Market Balance Area receiving the capacity (In Area) and the price of capacity in a Market Balance Area providing the capacity (Out Area), i.e. In Area Price - Out Area price.	A48
Inflow	The volume of water that flows into a reservoir in a given interval.	A49
Water extraction	The volume of water that can be extracted from a reservoir in a given interval.	A50
Turbined water	The volume of water that can be turbined in a plant in a given interval.	A51
Water spillage	The volume of water that is not turbined going through the spillway in a given interval.	A52
Planned maintenance	Maintenance has been planned for the object in question with a forecast ending date.	A53
Unplanned outage	An unplanned outage has occurred on the object in question.	A54
Use it Or Sell it (UIOSI) pricing	The time series provides information on the capacity resold in the "use it or sell it" process and its corresponding price.	A55
Compensation for auction cancellation where capacity is for resale	The time series provides information on the compensation of the capacity for resale following an auction cancellation.	A56
Resale pricing	For each Physical Transmission Rights holder, this document contains the resold capacity and its corresponding price.	A57
Curtailed capacity compensation	The time series provides information to compensate a party when curtailment is applied on the capacity obtained in a previous auction, resale or transfer.	A58
Use it Or Sell it (UIOSI) compensation	The time series provides information on the compensation for the capacity following an auction cancellation.	A59
Minimum possible	The time series provides a schedule of minimum possible values for a Resource Object. The nature of the flow could be defined by the attribute Direction.	A60
Maximum available	The time series provides a schedule of maximum available values for a Resource Object. The nature of the flow could be defined by the attribute Direction.	A61
Spot price	The time series provides the market spot prices from an auction.	A62
Minimum ATC	The Available Transmission Capacity that must be guaranteed because of regulatory constraints.	A63

Meter Measurement data	The data as provided for a meter measurement source.	A64
Accounting Point Relevant data	The metered data that is to be considered relevant for accounting purposes.	A65
Energy flow	Energy flow information.	A66
Power plant energy Schedule	Energy flow scheduled for a power plant.	A67
Compensation Requirements for the compensation period	The time series provides the compensation requirements for a given compensation period.	A68
Market coupling results	The time series provides the results of a market coupling auction.	A69
Production, unavailable	Production capacity that normally would be available, but due to maintenance or similar is temporarily unavailable.	A70
Supplementary available generation	The supplementary generation that is available.	A71
Interruptible consumption	The consumption that may be interrupted on request.	A72
Summarised Market Balance Area Schedule	A time series providing the total exchanges based on commercial transactions between two Market Balance Areas.	A73
Load Frequency Control Program Schedule	A time series providing the schedule information for the Load Frequency Control Program.	A74
Timeframe Independent Schedule	A time series providing the total exchanges of Timeframe Independent Schedules between two System Operators.	A75
Consumption curtailment	A time series providing the amount of voluntary consumption curtailed by the energy supplier of an end-consumer.	A76
Production, dispatchable	The nature of the business being described is dispatchable production details, i.e. generation output that can be changed by a request (activation order) of the TSO according with the applicable Market Rules.	A77
Consumption, dispatchable	The nature of the business being described is dispatchable consumption details, i.e. consumption output that can be changed by a request (activation order) of the TSO according with the applicable Market Rules.	A78
Production, non-dispatchable	The nature of the business being described is non-dispatchable production details, i.e. generation output that cannot be modified by an activation order.	A79
Consumption, non-dispatchable	The nature of the business being described is non-dispatchable consumption details, i.e. consumption output that cannot be modified by an activation order.	A80
Total Transfer Capacity (TTC)	The Total Transfer Capacity is the maximum exchange program between two areas compatible with operational security standards applicable at each system if future network conditions, generation and load patterns were perfectly known in advance.	A81
Mutual Emergency Assistance Service (MEAS)	The cross border Mutual Emergency Assistance Service between System Operators that are to be activated through schedule nomination.	A82
Auction cancelation	The time series covers auction cancellation right.	A83
Nomination curtailment	The time series covers nomination curtailment rights	A84
Internal redispatch	Redispatch to relieve Market Balance Area internal congestion.	A85
Control area balance energy	A sum of secondary, tertiary control as well as other energy that was used to balance a control area.	A86
Balancing energy price	Price of energy used to balance.	A87
Economised secondary reserve	The activated secondary reserve that had been economised due to pooled reserve management.	A88
Spinning reserve	The extra generating capacity that is available by increasing the production of generators that are already connected to the power system.	A89
Solar	The business being described concerns solar power.	A90
positive forecast margin	The business being described concerns a positive forecast margin.	A91
Negative forecast margin	The business being described concerns a negative forecast margin.	A92
Wind generation	The business being described concerns wind generation.	A93
Solar generation	The business being described concerns solar generation.	A94
Frequency		

containment reserve	The business being described concerns frequency containment reserve.	A95
Automatic frequency restoration reserve	The business being described concerns automatic frequency restoration reserve.	A96
Manual frequency restoration reserve	The business being described concerns manual frequency restoration reserve.	A97
Replacement reserve	The business being described concerns replacement reserve.	A98
Financial information	The business being described concerns financial information.	A99
Interconnector network evolution	The business being described concerns interconnector network evolution.	B01
Interconnector network dismantling	The business being described concerns interconnector network dismantling.	B02
Counter trade	The business being described concerns counter trades.	B03
Congestion costs	The business being described concerns congestion costs.	B04
Capacity allocated (including price)	The business being described concerns capacity allocation and includes price information.	B05
DC link constraint	The business being described concerns DC link constraints.	B06
Auction revenue	The business being described concerns auction revenue.	B07
Total nominated capacity	The business being described concerns the total nominated capacity.	B08
Net position	The business being described concerns net position.	B09
Congestion income	The business being described concerns congestion income.	B10
Production unit	The business being described concerns a production unit.	B11
Rounded market coupling results	Rounded outputs of the market coupling to be sent to TSOs and Market Participants.	B12
Allocation Revenue	The time series provides information on the revenue generated by the allocations.	B13
Production deviation	A time series concerning the imbalance energy between the metered production and the schedules calculated by an imbalance settlement responsible.	B14
Consumption deviation	A time series concerning the imbalance energy between metered consumption and the forecasted consumption calculated by an imbalance settlement responsible.	B15
Transmission asset	The business being described concerns a transmission asset.	B16
Consumption unit	The business being described concerns a consumption unit.	B17
In-feed ATC	Available Transfer Capacity at the in-feed side of a DC tieline.	B18
Out-feed ATC	Available Transfer Capacity at the out-feed side of a DC tieline.	B19
Balance up regulation price	A time series concerning balance regulation market prices for up regulation.	B20
Balance down regulation price	A time series concerning balance regulation market prices for down regulation.	B21
Main direction	A time series concerning the direction of balance regulations.	B22
Consumption imbalance price	A time series concerning imbalance prices for consumption.	B23
Production sales imbalance price	A time series concerning imbalance prices for production sales.	B24
Production purchase imbalance price	A time series concerning imbalance prices for production purchase.	B25
Average balance price between MBAs	A time series concerning the average prices between Market Balance Areas.	B26
Pumped	A time series concerning the electricity consumption related to pumping.	B27
Large installation consumption	A time series concerning consumption from large installation.	B28
Metering Grid Area (MGA) imbalance	A time series concerning imbalance between reported consumption, production and exchange in a Metering Grid Area.	B29
HVDC Link settings	The time series in question provides HVDC Link settings.	B30
Transmission Reliability Margin (TRM)	A time series concerning Transmission Reliability Margin (TRM).	B31
Imbalance component for a pool	This information is used to provide to a pool manager the combined imbalance of all the pool participants.	B32
Area Control error (ACE)	The sum of the instantaneous difference between the actual and the set-point value of the measured total power value and Control Program including Virtual Tie-Lines for the power interchange of a LFC Area or a LFC Block and the frequency bias given by the product of the	B33

Area Control Error	K-Factor of the LFC Area or the LFC Block and the Frequency Deviation.	
after Imbalance Netting	A time series concerning the Area Control Error after applying the imbalance netting energy correction.	B34
Implicit and explicit trade total	The sum of cross border schedules based on implicit and explicit trades including long term, yearly, monthly, weekly, daily processes.	B35
Production units own consumption	The consumption of one or more production units.	B36
Constraint situation	The timeseries describes the constraint situation for a given TimeInterval. A constraint situation can be: - composed of a list of network elements in outage associated for each outage to a list of network elements on which remedial actions have been carried out accordingly to contingency process - or it can be an external constraint.	B37
Initial domain	The timeseries describe the full flow based domain for a given TimeInterval. Critical network elements are displayed in details and their impact on the market is quantified.	B38
Flow based domain adjusted to long term schedules	The timeseries describe the full flow based domain for a given TimeInterval adjusted to the latest update of the schedules. Critical network elements are displayed in details and their impact on the market is quantified.	B39
Network element constraint	The timeSeries describes limiting elements which are overloaded.	B40
Calculation opposition (Red Flag)	The timeSeries describes a party who is opposed to the calculation result and imposes its transfer capacity value.	B41
Base case proportional shift key	The GSK or LSK are proportional to the base case generation or load.	B42
Proportional to participation factors shift key	The GSK or LSK are proportional to the participation factors.	B43
Proportional to the remaining capacity shift key	The GSK is proportional to the remaining available capacity.	B44
Merit order shift key	The GSK is proportional to a merit order list.	B45
Wind speed	The TimeSeries provides information on the wind speed.	B46
Wind direction	The TimeSeries provides information on the wind direction.	B47
Solar irradiance	The TimeSeries provides information on the power per unit area produced by the sun in the form of electromagnetic radiation.	B48
Air temperature	The TimeSeries provides information on the air temperature.	B49
Cloudiness	The TimeSeries provides information on the cloudiness, i.e. the level of coverage of the sky with clouds.	B50
Air humidity	The TimeSeries provides information on the level of humidity of the air.	B51
Atmospheric pressure	The TimeSeries provides information on the atmospheric pressure.	B52
Precipitation	The TimeSeries provides information on the amount of rain, snow, etc. that falls on the ground.	B53
Network constraint situation that constraints the market	The TimeSeries describes the network elements, that constraints the market, to be taken into account to simulate a network constraint during the network load flow studies. The network situation includes the contingencies, the remedial actions, the monitored network elements and the potential additional constraints.	B54
Contingency	The TimeSeries describes the network elements part of the contingency to be simulated for a given TimeInterval.	B55
Remedial Action	The TimeSeries describes a set of remedial actions for a given TimeInterval.	B56
Monitored Network Element	The TimeSeries describes the network elements to be monitored during the network load flow studies.	B57
Busbar	The TimeSeries describes the network elements that composed a busbar.	B58
Network Element	The TimeSeries describes network elements.	B59
SPS	The TimeSeries describes the network elements managed by a Special Protection System (automation).	B60
Aggregated netted external market schedule	The aggregated netted external market schedules for a given border.	B61
Aggregated netted external TSO schedule	The aggregated netted external TSO schedules for a given border.	B62

Aggregated netted external schedule	The aggregated netted external schedules for a given border.	B63
Netted area AC position	The AC position for a given area.	B64
Netted area position	The netted aggregation of all AC external schedules of an area plus the aggregated External Netted Schedules of related HVDC links of an area.	B65
Interconnection shift key	The shift key series describes the amount of power to be shifted from a border area.	B66
DC flow with losses	DC flow with losses refers to the values at the importing end of the DC line.	B67
DC flow without losses	DC flow without losses refers to the values at the exporting end of the DC line.	B68
minimum value of netted area position	That value which a netted area position must not fall below for a given area.	B69
maximum value of netted area position	That value which a netted area position must not exceed for a given optimisation area.	B70
maximum value of DC flow	That value which a balanced DC flow must not exceed for a given DC line on exporting end. When aligning DC flows CGMA algorithm will respect this constraint.	B71
minimum value of DC flow	That value which a balanced DC flow must not fall below for a given DC line on exporting end. Currently this business type is only included for consistency reasons. It is always set to 0. This constraint might, however, be used in future. When aligning DC flows the CGMA algorithm will respect this constraint.	B72
indicative AC flow	It is the hypothetical flow on the aggregate of all AC tie lines of an electrical border between two optimisation areas. It results from the adjustments to the preliminary netted area positions of all optimisation areas made by the CGMA algorithm. Indicative AC flows are an artefact of the CGMA algorithm, and do not correspond to physical flows	B73
Offer	The time series provides an offer to provide reserves.	B74
Need	The time series provides a requirement for reserves.	B75
Opportunity costs or benefits	The time series describes any opportunity costs or benefits.	B76
Financial compensation or penalties	The time series describes any financial compensation or penalties	B77
Global radiation	The total short-wave radiation from the Global radiation is the total short-wave radiation from the sky falling onto a horizontal surface on the ground. It includes both the direct solar radiation and the diffuse radiation resulting from reflected or scattered sunlight.	B78
Diffuse radiation	Radiation resulting from reflected or scattered sunlight.	B79
Direct solar radiation	Radiation resulting from direct sunlight	B80
Outage (OUT)	Outage process: Element is out of operation due to planned maintenance or due to an unplanned/forced outage. Outage may be used as a synonym for unavailability.	B81
Special switching state (SSS)	Outage Process: This state applies to grid elements which are in operation in an exceptional state (e.g. separated nodes operation).	B82
Testing (TEST)	Outage process: TESTING means any element status is possible - ON or OUT. This status applies either between first connection and final commissioning of the relevant asset, or directly following maintenance of the relevant asset.	B83
Auxiliary busbar operation	Outage process: Element is in operation but connected via auxiliary busbar	B84
Automatic reclosing	Outage process: Protection function Automatic reclosing is switched off for electric line	B85
Busbar protection	Protection function busbar protection is switched off	B86
Phase Shift Angle	The maximum phase shift angle allowed between two network elements.	B87
Base Case Network Situation	The TimeSeries describes the network elements to be taken into account to simulate a base case network situation during the network load flow studies, without any contingency.	B88
Inter-TSO assistance	Cross border assistance schedule between TSOs not interconnected directly.	B89
FlexibleNeed	The business type indicates that the need is optional.	B90
GLSK Limitation	A constraint related to a GLSK maximum or minimum limitation in the production or/and consumption shift.	B91
Capacity ramping limitation	A constraint related to a ramping limitation on the capacity offered at a given border.	B92
interconnector capacity	The maximum capacity that can be exchanged on an interconnector, excluding external factor on both ends.	B93
Must Run	A time series concerning must run generation.	B94
Procured capacity	An accepted offer of balancing capacity.	B95

Used capacity	The used cross-zonal balancing capacity.	B96
Estimated costs	Estimated costs of the process.	B97
Estimated benefits	Estimated benefits of the process.	B98
Load Shedding	A time series concerning a load shedding used to avoid failure of the power system.	B99
Remaining Capacity	A time series concerning the remaining capacity.	C01
Indicator of generation capacity adequacy	Indicator of adequacy, it indicates if there is final generation remaining capacity after SMTA calculation.	C02
Income from price divergence without congestions	The time series describes income due to price divergence without congestion between bidding zones.	C03
Push-button	The cross-border Push-button service between System Operators.	C04
Intertripping	The cross-border Intertripping service between System Operators.	C05
Emergency instruction	The cross-border Emergency instruction service between System Operators.	C06
Ramp management	The schedule resulting from cross-border Ramp management service between System Operators.	C07
Profile smoothing	The schedule resulting from cross-border Profile smoothing service between System Operators.	C08
Emergency reallocation deselection	The schedule resulting from cross-border Emergency reallocation deselection service between System Operators.	C09
SO-SO-trade	The generic cross border trade between System Operators.	C10
Production reduction	A time series providing the volume of production reduced by an energy provider / producer / supplier.	C11
Maximum power exchange	The timeseries provides the maximum admissible power flow between two bidding zones respecting operational security limits taking into account N-1 criterion.	C12
Maximum power exchange after remedial actions	The timeseries provides the maximum admissible power flow between two bidding zones after remedial actions.	C13
Network constraint situation that cannot limit the market	The TimeSeries describes the network elements, that cannot limit the market, to be taken into account to simulate a network constraint during the network load flow studies. The network situation includes the contingencies, the remedial actions, the monitored network elements and the potential additional constraints.	C14
Flat participation for all generators or loads	Flat GSK factors of all generators or loads, independently of the size.	C15
Proportional to installed capacity of generators	Generators participate relative to their maximum (installed) capacity (MW).	C16
Market price and total volume	A time series concerning market price and total volume.	C17
Import price	A time series concerning import price (the volume-weighted price average of all accepted bids).	C18
Capacity allocated (excluding price)	The business being described concerns capacity allocation and excludes price information.	C19
Common Grid Model Equipment	The timeseries provides equipment related to the Common Grid Model (CGM).	C20
Exchanged balancing reserve capacity	The balancing reserve capacity exchanged between areas.	C21
Shared balancing reserve capacity	The balancing reserve capacity shared between areas.	C22
Share of reserve capacity	A time series concerning the share of reserve capacity.	C23
Actual reserve capacity	A timeseries concerning actual reserve capacity.	C24
K-factor Frequency	K-factor as stated in the SO GL Art. 2 (45). It is also known as Frequency Bias.	C25
Containment Reserve-Normal (FCR-N)	FCR-N is a reserve that is automatically activated in both directions around a set point when the frequency varies between 50.10 Hz and 49.90 Hz after an imbalance.	C26
Frequency Containment Reserve-Disturbance (FCR-D)	FCR-D is a reserve that is automatically activated when the frequency falls below 49.90 Hz or rises above 50.1 Hz after an imbalance.	C27
Internal trade difference	A time series concerning internal trade difference, within an area, such as a Bidding Zone or Scheduling Area. The internal trade difference is the difference between trades reported from an out party (seller) and an in party (buyer).	C28

Small scale production	Production from small scale production plants.	C29
System price	The system price is an unconstrained market clearing reference price. It is calculated without any congestion restrictions by setting capacities to infinity.	C30
Wind gust	An increase in the speed of the wind lasting for a short period.	C31
Area imbalance	A time series concerning imbalance between planned consumption, production and exchange in an Area.	C32
Unintended energy price	A timeseries concerning the price of the unintended cross-border exchange of energy.	C33
Frequency containment process energy	A timeseries containing the volume of energy resulting from the frequency containment process.	C34
Frequency containment process energy price	A timeseries containing the energy price from the frequency containment process.	C35
Ramping period energy	A timeseries containing the volume of energy exchanged as a result of ramping between different ANES values.	C36
Ramping period energy price	A timeseries containing the price of the energy exchanged as a result of ramping between different ANES values.	C37
Frequency deviation	A timeseries concerning the difference between the actual and the nominal frequency of a synchronous area.	C38
Day-Ahead market price	A timeseries concerning Day-Ahead market prices.	C39
Conditional bid	Standard product bid that is conditional on bids submitted outside of common platform.	C40
Thermal limit	The current causing a given network element to work outside of the range of safe operating temperatures.	C41
Frequency Limit	A constraint related to the containment of frequency deviations within a given area.	C42
Voltage limit	The maximum or minimum permissible voltage within normal operation state of a given network element.	C43
Current limit	The maximum permissible current within normal operation state of a given network element.	C44
Short circuit current limit	The maximum permissible short-circuit current within normal operation state of a given network element.	C45
Dynamic stability limit	A maximum permissible load ensuring the control of oscillations in the grid and avoiding the loss of synchronism.	C46
Disconnection	A timeseries describing disconnection of a TSO from a common platform.	C47
Intended energy with positive price	A timeseries concerning the amount of intended energy with prices higher than zero (and including zero).	C48
Intended energy with negative price	A timeseries concerning the amount of intended energy with prices lower than zero (excluding zero).	C49
Decopuling	A time series describing decoupling of an area.	C50
Resource capacity unit	A timeseries containing information about resource capacity units.	C51
Resource entry capacity data	A timeseries containing the resource capacity that can be allocated to an eligible resource capacity operator from another area.	C52
Resource capacity obligation data	A timeseries containing the resource capacity operator obligation to guarantee delivery.	C53
Available energy	A timeseries concerning the available energy.	C54
Production curtailment	A timeseries concerning the curtailment of production.	C55
Rounding error	A timeseries describing a rounding error.	C56
Metered frequency	The timeseries provides information about metered frequency.	C57
Adjusted TTC to the nominal criteria	The exchange program between two areas which guarantees that the Margin Available for Cross-Zonal Trade (MACZT) fulfils the nominal criteria at least on the most limiting Critical Network Element with Contingency (CNEC) which limits the transfer capacity.	C58
Adjusted TTC to the nominal criteria with TSOs limitation	The exchange program between two areas which allows the Margin Available for Cross-Zonal Trade (MACZT) on, at least, the most limiting Critical Network Element with Contingency (CNEC) to get closer to the nominal criteria fulfilment with a limited impact on the rest of the network.	C59
Frequency deviation larger than standard deviation	Total time in which the absolute value of the instantaneous frequency deviation was larger than the standard frequency deviation.	C60
Frequency deviation larger	Total time in which the absolute value of the instantaneous frequency deviation was larger	

than maximum deviation	than the maximum instantaneous frequency deviation.	C61
Frequency deviation not returned to 50%	Number of events in which the absolute value of the instantaneous frequency deviation of the synchronous area exceeded 200 % of the standard frequency deviation as stated in SO GL (EU) regulation Art 131.1.a.vi.	C62
Frequency deviation not returned to restoration range	Number of events in which the absolute value of the instantaneous frequency deviation of the synchronous area exceeded 200 % of the standard frequency deviation.	C63
Frequency deviation outside recovery range	Number of events for which the absolute value of the instantaneous frequency deviation was outside of the frequency recovery range.	C64
Frequency	A time series describing measurement frequency.	C65
Mean value	A time series describing mean values.	C66
Standard deviation	A time series describing standard deviation.	C67
Percentile	A time series describing percentiles.	C68
Measured frequency resolution	A time series describing the resolution of a measured frequency.	C69
Accuracy	A time series describing measurement accuracy.	C70
FRCE outside level 1 range	The number of time intervals in which the average value of the FRCE was outside the Level 1 FRCE range as stated in SO GL. (EU) regulation Art 131.1.b.i.	C71
FRCE outside level 2 range	The number of time intervals in which the average value of the FRCE was outside the Level 2 FRCE range as stated in SO GL (EU) regulation. Art 131.1.b.i.	C72
FRCE exceeded 60% of FRR capacity	The number of events for which the FRCE exceeded 60 % of the reserve capacity on FRR as stated in SO GL (EU) regulation Art 131.1.b.ii.	C73
FRCE exceeded steady state deviation	The number of events for which the absolute value of the FRCE exceeded the maximum steady-state frequency deviation.	C74
FRCE calculation and accuracy descriptor	A time series describing how FRCE is calculated and its accuracy.	C75
Forecasted capacity	A time series describing forecasted capacity.	C76
Minimum available capacity	A time series describing minimum available capacity.	C77
Average available capacity	A time series describing average available capacity.	C78
Maximum available capacity	A time series describing maximum available capacity.	C79
Frequency and accuracy descriptor	A time series describing how system frequency and accuracy are determined.	C80
Long-Term internal redispatch	A time series describing long-term redispatch to relieve Scheduling Area internal congestion.	C81
Other unavailability	This is an unplanned unavailability. Not considered by market participants as a planned maintenance.	C82
Faster than standard FAT	Bids that can support a ?Full Activation Time? (FAT) that is faster than standard FAT. Fast activation can be done for bids with activation time shorter than the minimum requirement for the standard product. When circumstances call for it, the TSO can order activation of such bids on a shorter notice.	C83
Faster than standard deactivation time	Fast deactivation can be done for bids with activation time shorter than the minimum requirement for the standard product. When circumstances call for it, the TSO can order activation of such bids on a shorter notice.	C84
Slower than standard FAT	Bids that can support a Full Activation Time (FAT) that is slower than standard FAT.	C85
Remedial action cost summary	A timeseries summarizing all incurred costs and/or revenues per party related to the activated remedial actions eligible to Cost Sharing.	C86
Settlement result	A timeseries representing the cost sharing settlement results between parties (i.e. which party will be paying and/or receiving money for the concerned period).	C87
Reserved cross zonal capacity	The reserved cross zonal capacity for system operator needs (EBGL art. 38(1)(b) and art. 41.).	C88
Energy reserves	A timeseries describing energy reserves.	C89
Combined dynamic constraint	The time series describes a combined dynamic constraint, being defined as a limit on the sum of power flows on a set of network elements or partial flows on a set of network elements for the purpose to respect dynamic stability limits.	C90
Anonymised constraint	The time series describes an anonymised constraint.	C91

Synchronous condenser activation	Activates a number of generators or other units in synchronous condenser mode.	C92
Permission administration details	A timeseries describing permission details such as authorisations, licenses or related regulatory requirements.	C93
Maximum capacity value of AC border flow	Maximum flow on an AC border between two areasÂ (capacity)	C94
Initial forecast value of AC border flow	Initial flow on an AC border between two areas (forecast)	C95

EnergyProductTypeList

The identification of the nature of an energy product such as power, energy, reactive power, etc.

Literal	Description	Code
Active power	The product of voltage and the in-phase component of alternating current measured in units of watts and standard multiples thereof.	8716867000016
Reactive power	The product of voltage and current and the sine of the phase angle between them, measured in units of voltamperes reactive and standard multiples thereof. (not used for planned schedules).	8716867000023
Active energy	The electrical energy produced, flowing or supplied by an electrical circuit during a time interval, being the integral with respect to time of instantaneous active power, measured in units of watt-hours, or standard multiples thereof.	8716867000030
Reactive energy	The integral with respect to time of reactive power (not used for planned schedules).	8716867000047
Capacitive Reactive energy	Capacitive reactive energy.	8716867000139
Inductive Reactive energy	Inductive reactive energy.	8716867000146
Water	For hydro power stations, this enables the identification of the quantity of water stored behind a dam (volume, head level, etc.), or the constraints in the flow of water.	8716867009911
Capacitive reactive power	Capacitive reactive power.	8716867000115
Inductive reactive power	Inductive reactive power.	8716867000122

ObjectAggregationTypeList

The identification of the domain that is the common dominator used to aggregate a time series.

Literal	Description	Code
Area	The object being described concerns an area.	A01
Metering point	The object being described concerns a metering point.	A02
Party	The object being described concerns a party.	A03
Agreement Identification	The object being described concerns an agreement identification.	A04
Accounting point	The object being described concerns an accounting point.	A05
Resource Object	The object being described concerns a resource object.	A06
Tieline	The object being described concerns a tieline.	A07
Resource type	The object being described concerns a resource type.	A08
DC link	The object being described concerns a DC link.	A09
AC link	The object being described concerns an AC link.	A10
Merchant line	The object being described concerns a merchant line.	A11
Bidding zone	The largest geographical area within which market participants are able to exchange energy without capacity allocation.	A12
Virtual bidding zone	A non-geographical bidding zone to be able to apply extra constraints to Bidding Zones.	A13
Metering grid area	A physical area where consumption, production and exchange can be measured. It is delimited by the placement of meters for continuous measurement for input to, and withdrawal from the area.	A14

Scheduling area	An area within which the TSOs' obligations regarding scheduling apply due to operational or organisational needs.	A15
Exchange point	The object being described concerns an exchange point.	A16

CurveTypeList

The type of curve being defined in the time series.

Literal	Description	Code
Sequential fixed size block	The curve is made of successive Intervals of time (Blocks) of constant duration (size), where the size of the Blocks is equal to the Resolution of the Period.	A01
Point	The curve is made of successive instants of time (Points).	A02
Variable sized Block	The curve is made of successive Intervals of time (Blocks) of variable duration (size), where the end date and end time of each Block are equal to the start date and start time of the next Interval. For the last Block the end date and end time of the last Interval would be equal to EndDateTime of TimeInterval.	A03
Overlapping breakpoint	The curve is made of successive Intervals of time of variable duration (size), where the end date and end time of each interval are equal to the start date and start time of the next Interval.	A04
Non-overlapping breakpoint	This curve is a restriction of the curve type A04, i.e. overlapping breakpoints. The restriction is that a single Period is allowed.	A05

Compounds

ESMP_DateTimeInterval

This datatype enables to express the start date and time, and the end date and time of a time interval with a specific pattern. This pattern is the YYYY-MM-DDThh:mmZ.

!# shows all attributes of ESMP_DateTimeInterval.

Table !# - Attributes of ESMPDataTypes::ESMP_DateTimeInterval.

Attribute name	mult.	Attribute type	Description
start (Compound)	1..1	YMDHM_DateTime	The start date and time of the interval with a minute resolution.
end (Compound)	1..1	YMDHM_DateTime	The end date and time of the interval with a minute resolution.

Primitives

String

A string consisting of a sequence of 8 bit characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

Integer

An integer number. The range is unspecified and not limited.

Decimal

Decimal is the base-10 notational system for representing real numbers.

DateTime

Date and time as "YYYY-MM-DDThh:mm:ss.sssZ", which conforms with ISO 8601 UTC time zone.
Note: the time within ESMP is expressed in UTC.

Duration

Duration as "PnYnMnDTnHnMnS" which conforms to ISO 8601, where nY expresses a number of years, nM a number of months, nD a number of days. The letter T separates the date expression from the time expression and, after it, nH identifies a number of hours, nM a number of minutes and nS a number of seconds. The number of seconds could be expressed as a

decimal number, but all other numbers are integers.